

ALBERT-LUDWIGS-UNIVERSITÄT FREIBURG

Master's Thesis

**Benchmarking Deep Learning Frameworks for
Individual Tree Segmentation and Species
Classification in Temperate Forests**

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Abstract

Individual-tree inventories remain labour-intensive, while deep learning on terrestrial laser scanning (TLS) promises scalable instance segmentation and species classification. Prior TLS instance-segmentation benchmarks already show that performance varies across contrasting forest conditions. Building on that landscape, this thesis jointly evaluates segmentation and species classification on temperate mixed TLS scenes, with the strongest contribution being which trees and neighbourhoods remain challenging for current models. It builds manually refined TLS instance references for two University of Freiburg research forests in southwestern Germany: a beech-dominated Ettenheim plot (approximately 50 m $\times$ 50 m; 128 ground-truth trees) and a denser spruce-dominated Mathislewald subplot (approximately 25 m $\times$ 25 m; 189 ground-truth trees). ForestMamba and SegmentAnyTree are compared against that geometric reference using point-count IoU matching. Inventory stems are linked exclusively to manual trees, and DetailView is applied to the same matched stems under manual, SegmentAnyTree, ForestMamba and ForestMamba+BluePoint instance tracks. Exploratory per-tree associations (trees within subplots are not independent replicates) separate two site-specific failure regimes for hard F1: at Ettenheim, nearest-neighbour packing among whole-tree centroids dominates, with crowded trees losing roughly 22 percentage points of ForestMamba hard F1 and 45 points of SegmentAnyTree hard F1 relative to the most open tercile. At Mathislewald, height and whole-tree convex-hull volume dominate, and both models miss most small, low-volume reference trees (lowest size terciles near 0–4% hard F1). Plot-level F1 at IoU threshold 0.5 under this manually refined reference ranked ForestMamba ahead on both sites (70.3% at Ettenheim; 37.0% at Mathislewald), with SegmentAnyTree competitive at Ettenheim (64.2%) but weaker at Mathislewald in the final absolute-coordinate comparison (14.5%). Those rankings are conditional on a reference edited from preliminary ForestMamba instances. On inventory-matched stems, DetailView overall accuracy on manual instances was 46.8% at Ettenheim ($N = 111$) and 62.5% at Mathislewald ($N = 32$). The Mathislewald figure is sample-limited and is limited mainly by open-set taxonomic transfer and class imbalance rather than by delineation quality. Free space among whole-tree centroids showed no clear evidence of improving species accuracy, and large whole-tree hull volume at Ettenheim tracked poorer labels through majority beech failure, so structural covariates useful for segmentation are not interchangeable predictors of taxonomic correctness. Switching to automatic instances changed stem-level overall accuracy by only a few percentage points at Ettenheim and by noisier, sample-limited amounts at Mathislewald. This ablation measures classification given matched instances, not end-to-end species-map accuracy. The study contributes a reproducible local reference protocol and evidence that improving instances or packing-based triage alone is unlikely to close the taxonomic gap without local adaptation.

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List of abbreviations

Abbreviations and short forms used in this thesis are defined below. Model and product names that are written in full throughout (for example ForestMamba, SegmentAnyTree and DetailView) are omitted.

- **ALS**: Airborne laser scanning
- **CCB2**: Modern CloudCompare binary project format
- **CI**: Confidence interval
- **CRS**: Coordinate reference system
- **DBH**: Diameter at breast height
- **ECOSENSE**: Ecosystem sensing research infrastructure / forest at Ettenheim
- **EPSG**: Coordinate reference system code registry (here EPSG:32632, UTM zone 32N)
- **F1**: Harmonic mean of precision and recall
- **FN**: False negative (unmatched ground-truth tree at the IoU gate)
- **FOR-species20K**: Large-scale forest tree-species classification benchmark used by DetailView
- **FP**: False positive (unmatched predicted instance at the IoU gate)
- **GT**: Ground truth (manual instance reference)
- **IoU**: Intersection over Union (point-count overlap used for instance matching)
- **LAZ**: Compressed ASPRS LAS point-cloud file format
- **MLS**: Mobile laser scanning
- N : Sample size (trees or stems in the stated set)
- N_{GT} : Number of ground-truth tree instances
- N_{pred} : Number of predicted tree instances
- **NN**: Nearest neighbour (planimetric distance between whole-tree centroids of manual ground-truth trees; not breast-height stem spacing)
- **Hull volume**: Whole-tree convex-hull volume (3D convex hull of all retained points of a manual tree; not crown-only volume)
- **OA**: Overall accuracy
- **PredInstance**: Stem / instance identifier field used for DetailView packages

- **QA**: Quality assurance
- ρ : Spearman rank correlation coefficient
- τ : IoU matching threshold
- **TLS**: Terrestrial laser scanning
- **TP**: True positive (accepted ground-truth–prediction match)
- **ULS**: Unmanned / UAV laser scanning
- X_{eval} : Shared evaluation point set used for instance scoring
- **XYZ**: Cartesian spatial coordinates

1 Introduction

Knowing which trees grow where, and in what mixture, size and condition, is central to forest science. Tree lists that carry composition and spatial arrangement determine the quality of biomass estimates, habitat assessments, growth and yield projections, and evaluations of management interventions. Obtaining that individual-tree information remains a limiting step: field campaigns are time-intensive, costly, constrained by access, and difficult to sustain at the spatial and temporal density that modern questions demand (Bundesministerium für Ernährung und Landwirtschaft (BMEL), 2021; Christin et al., 2019; Mustafa and Habeeb, 2014; Smith, 2002).

Remote sensing, especially terrestrial laser scanning (TLS) combined with deep learning, is presently the most viable route toward inventories that reduce field labour while retaining stem-level geometric and taxonomic detail (Béland et al., 2019; Calders et al., 2020; Kattenborn et al., 2021; Liang et al., 2016; Lines et al., 2022; Newnham et al., 2015; White et al., 2016; Zhao et al., 2023). Much remains to be understood about how emerging sensors and models map onto individual-tree delineation, species identity, and the propagation of error along an automated workflow (Cherlet et al., 2026; Fassnacht et al., 2016; Hamedianfar et al., 2022; Pu, 2021). Benchmarking against carefully curated reference forests, with transparent metrics and documented failure modes across ecological contexts, is the mechanism that makes such contributions transferable (Cherlet et al., 2026; Henrich et al., 2024; Puliti et al., 2025, 2023; Xiang et al., 2024, 2025).

This thesis develops high-quality reference data for two university research forests in southwestern Germany and uses them to benchmark contemporary deep-learning workflows for individual-tree analysis. Recent TLS instance-segmentation benchmarks already show large performance spreads across methods and contrasting forest conditions (Cherlet et al., 2026). The present contribution is more specific: we jointly examine overall recovery of trees and species labels, how performance varies with tree height, whole-tree hull volume and neighbourhood, and how automatic instance source changes stem-level species labels relative to a manual baseline, while documenting a reproducible local protocol that others can extend.

1.1 Research questions

1. How accurately do ForestMamba and SegmentAnyTree recover manually verified individual trees in complex temperate TLS scenes, and how does DetailView perform on inventory-matched stems under manual instances?
2. How does performance vary with tree height ($z_{99} - z_{05}$), whole-tree convex-hull volume, and neighbourhood (planimetric nearest-neighbour distance among whole-tree centroids)?
3. To what extent do automatic instance sources change stem-level species labels relative to the manual-instance baseline on the same inventory-matched trees?

1.2 Objectives

Correspondingly, we aim to (i) curate manually refined TLS instance references co-registered with university field inventory where applicable; (ii) benchmark overall instance-segmentation and manual-track species performance against that reference using standardized metrics; (iii) quantify structural associations of per-tree hard F1 and DetailView correctness with height, whole-tree hull volume and nearest-neighbour packing; and (iv) quantify segmentation-to-species propagation across instance tracks and synthesize strengths, limitations and implications for a transferable local protocol.

2 Related work

2.1 Individual-tree instance segmentation

Individual-tree detection and delineation have long been central to remote-sensing inventories, first mainly with airborne laser scanning (ALS) crown products (Hyypä et al., 2008; Wang et al., 2016) and increasingly with dense proximal clouds from terrestrial, mobile and unmanned platforms (Béland et al., 2019; Guimarães et al., 2020; Liang et al., 2016; Newnham et al., 2015; White et al., 2016; Wilkes et al., 2017). Classical geometric pipelines reconstruct stems and crowns from TLS by region growing, topology or quantitative structure models (QSMs) (Burt et al., 2019; Hackenberg et al., 2015; Lowe and Stepanas, 2021; Raunonen et al., 2013).

Deep learning has shifted the field toward end-to-end instance and panoptic segmentation on LiDAR point clouds. SegmentAnyTree provides a sensor- and platform-agnostic pipeline transferable across ULS, TLS and MLS (Wielgosz et al., 2024). ForAINet couples semantic and instance heads for automated inventory attributes on high-density airborne clouds (Xiang et al., 2024). ForestFormer3D unifies semantic and instance segmentation with standardized detection metrics on expanded FOR-instance benchmarks (Puliti et al., 2023; Xiang et al., 2025). ForestMamba introduces a structure-aware sparse Mamba backbone with geometry-guided queries for forest scenes (Nguyen et al., 2026). TreeLearn shows that data-driven instance segmentation on ground-based clouds can match or exceed classical pipelines when high-quality labels are available (Henrich et al., 2024). Broader TLS instance-segmentation benchmarks highlight large performance spreads across methods and sites and underscore the need for transparent reference protocols (Cherlet et al., 2026).

2.2 Tree species classification from proximal LiDAR

Species classification from remote sensing has a long optical and multi-sensor literature, with persistent challenges for large-area, operational mapping (Fassnacht et al., 2016; Mustafa and Habeeb, 2014; Pu, 2021). From single-tree point clouds the field has moved from hand-crafted features toward deep multi-view and point-based networks. The FOR-species20K dataset and competition established an open baseline across 33 species and TLS/MLS/ULS modalities. DetailView (multi-view DenseNet) led the leaderboard (Puliti et al., 2025). A recurring limitation, directly relevant here, is that published classifiers often assume high-quality manual instances, and operational clouds from automatic segmenters may degrade species accuracy (Fassnacht

et al., 2016; Puliti et al., 2025).

2.3 Structural attributes, biomass and inventory linkage

Once trees are delineated, TLS supports plot-scale structural measurement including height, DBH and crown geometry (Cavalheiro et al., 2026; Jennings et al., 1999; Kükenbrink et al., 2022; Liu et al., 2018; Zhang et al., 2024). QSMs and related reconstructions enable nondestructive biomass estimation that can outperform purely allometric routes in some settings (Calders et al., 2015; Disney et al., 2018; Pourdelan et al., 2026; Raunonen et al., 2013; Riedel and Kändler, 2017; Tian et al., 2023). National inventory protocols remain the taxonomic and sampling backbone against which remote-sensing products are judged (Bundesministerium für Ernährung und Landwirtschaft (BMEL), 2021; Smith, 2002).

2.4 Reference data and evaluation gaps

Standardized national inventory protocols define field practice (Bundesministerium für Ernährung und Landwirtschaft (BMEL), 2021), but TLS deep-learning studies often rely on small, site-specific or incompletely documented labels (Cherlet et al., 2026; Hamedianfar et al., 2022; Puliti et al., 2023). Benchmark datasets such as FOR-instance improve comparability for segmentation (Puliti et al., 2023; Xiang et al., 2024, 2025). Cherlet et al. already show that instance-segmentation accuracy varies strongly across contrasting forest conditions (Cherlet et al., 2026), so forest complexity itself is not an unexplored axis. Building on that evidence, temperate mixed university forests with paired inventory and manually refined TLS instances remain useful local testbeds. Our contribution is to treat university field inventory as taxonomic ground truth and manually refined TLS instances as geometric ground truth, then jointly examine how tree size and local neighbourhood relate to segmentation and species-classification performance under a common footprint, including an explicit segmentation-to-species ablation.

3 Study sites and data

3.1 Study sites

This thesis is based on two temperate research forests in Baden-Württemberg, southwestern Germany. At each site field inventory and terrestrial laser scanning (TLS) were collected over the same stand footprint.

3.1.1 Ettenheim (ECOSENSE)

The Ettenheim site lies in the western foothills of the Black Forest near the town of Ettenheim, on the eastern margin of the Upper Rhine Plain (approximately 48.2680°N, 7.8782°E). Elevation is comparatively low for the wider Black Forest region, and the local climate is milder and warmer than that of the highlands to the southeast, with a temperate, seasonally distinct regime typical of the Upper Rhine / western Black Forest fringe.

The research plot analysed here measures approximately 50 m × 50 m (2500 m²). Within this footprint the field inventory records 115 stems. Composition is strongly beech-dominated:

European beech (*Fagus sylvatica*) accounts for 71.3% of inventoried stems, followed by silver fir (*Abies alba*, 15.7%) and Douglas fir (*Pseudotsuga menziesii*, 11.3%), with Norway spruce (*Picea abies*) and European larch (*Larix decidua*) each below 1%.

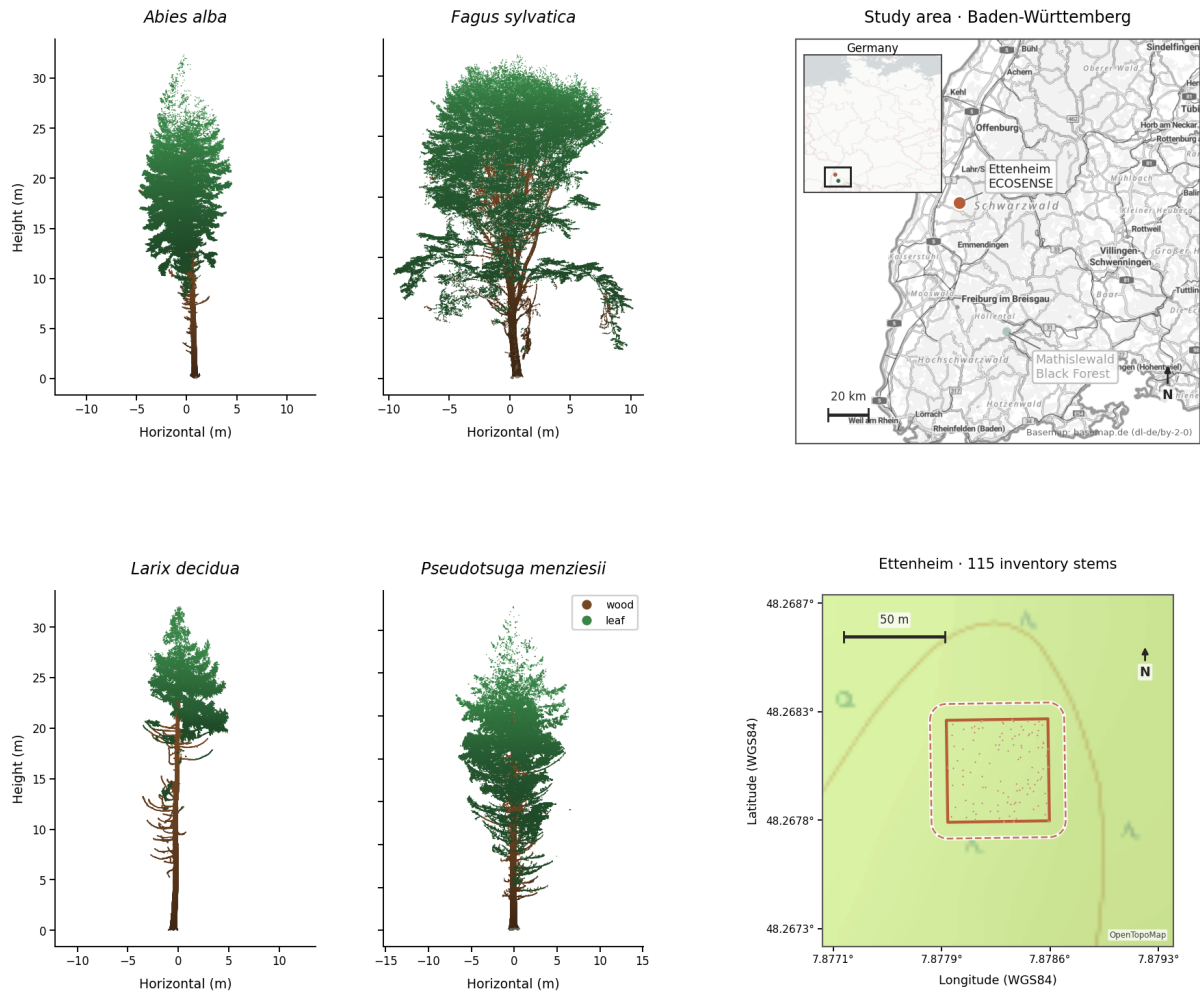


Figure 1: Ettenheim research plot. Left: coloured 2D side profiles of the largest inventory-matched tree of each species present in the plot (wood brown, foliage green). Right: location of the plot on a topographic map, with the approximate length and width of the research footprint indicated.

3.1.2 Mathislewald (Hinterzarten)

The Mathislewald site lies in the southern Black Forest near Hinterzarten, adjacent to Mathisleweiher (approximately 47.8847°N, 8.0885°E). It sits substantially higher than Ettenheim and experiences a cooler, more humid montane climate typical of the Black Forest uplands, with cooler summers, colder winters, and higher precipitation than the Upper Rhine fringe.

The research plot analysed here measures approximately 25 m × 25 m (625 m²). Within this footprint the field inventory records 34 stems. Composition is spruce-dominated: Norway spruce (*Picea abies*) accounts for 64.7% of inventoried stems, followed by European beech (*Fagus sylvatica*, 23.5%) and silver fir (*Abies alba*, 11.8%).

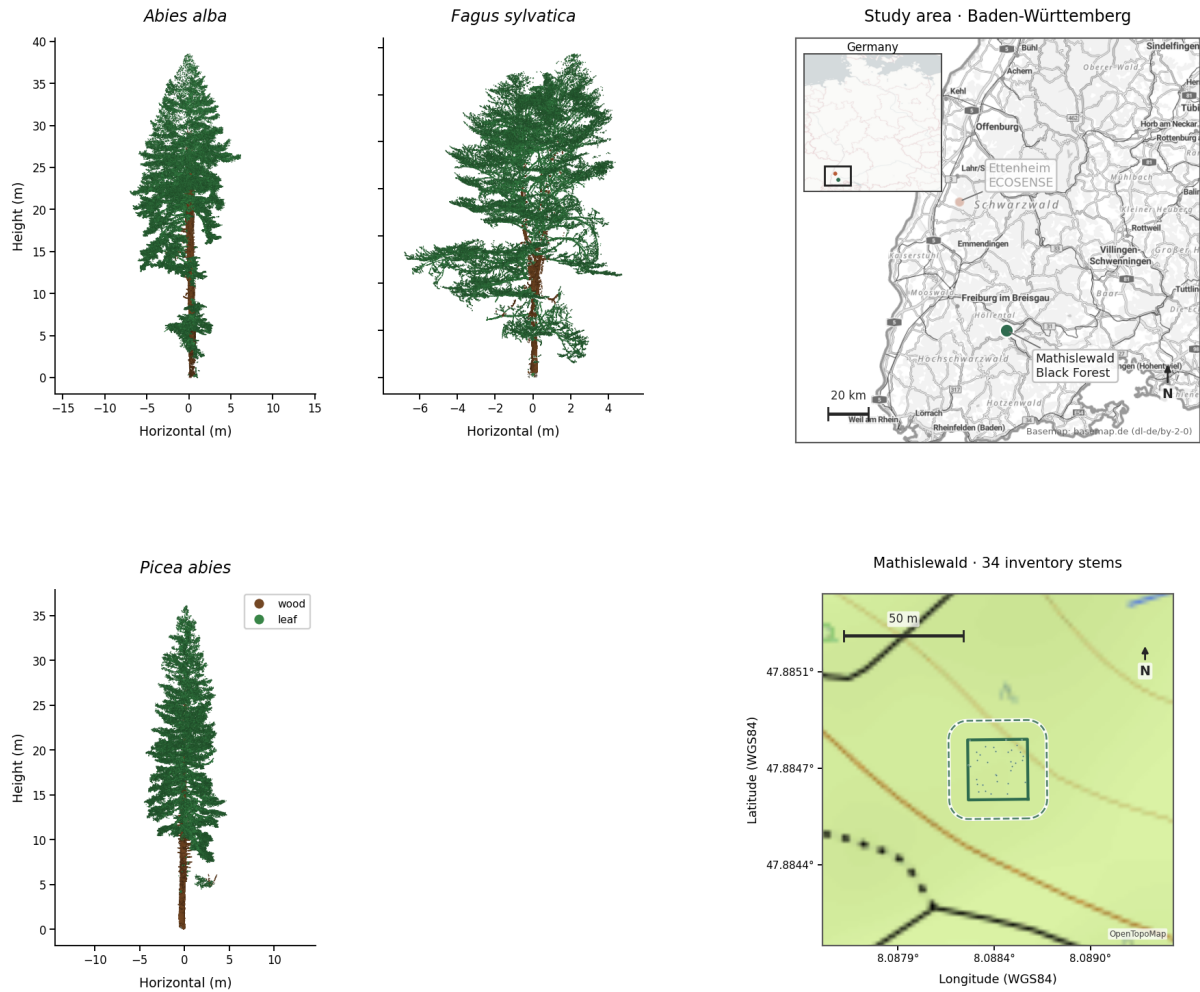


Figure 2: Mathislewald research plot. Left: coloured 2D side profiles of the largest inventory-matched tree of each species present in the plot (wood brown, foliage green). Right: location of the plot on a topographic map, with the approximate length and width of the research footprint indicated.

3.2 Project context

The Ettenheim research plot analysed here is part of the ECOSENSE forest site at the University of Freiburg, which develops advanced sensor networks for climate-threatened forest ecosystems and combines close-range remote sensing, vertical optical depth measurements, drone-based laser scanning and satellite time series (Chair of Sensor-based Geoinformatics, University of Freiburg, 2026; ECOSENSE - SFB 1537, University of Freiburg, 2026). The ECOSENSE forest lies in the municipal forest of Ettenheim and comprises a mixed system dominated by European beech with patches of spruce, oak, fir, Douglas fir and pine. Selected stands support site infrastructure such as eddy covariance and canopy access (ECOSSENSE - SFB 1537, University of Freiburg, 2026).

Mathislewald provides the University of Freiburg with a 127 ha research and teaching forest near Hinterzarten that is digitized with remote sensing methods and named as a pilot region in related digital-twin and AR developments (Professur für Waldwachstum und Dendroökologie, University of Freiburg, 2026; Wissen über den Wald im Wandel (WWW 4.0), University of Freiburg, 2026). That context aligns with WaldAgil, which targets forest digitization at the level of individual trees and supports digital-twin based scenario planning with drones, laser scanning, AI-based

tree recognition and augmented-reality interfaces (WaldAgil, University of Freiburg, 2026a,b).

3.3 Data and models

The field inventory provides expert taxonomic and structural attributes for stems. Traditional field measurement, including species determination and breast-height diameter assessment, is labour intensive, while laser scanning supports extraction of tree height, position, stem form and crown dimensions (Cavalheiro et al., 2026; Kükenbrink et al., 2022; Liang et al., 2016; Liu et al., 2018; WaldAgil, University of Freiburg, 2026b). Single-tree biomass and carbon accounting in German greenhouse-gas reporting likewise rest on dendrometric stem attributes (Calders et al., 2015; Riedel and Kändler, 2017; Tian et al., 2023).

Point clouds used in this thesis were collected as TLS and form the structural basis for manual instance reference building. Although drone-based laser scanning is part of the broader ECOSENSE and WaldAgil sensing portfolios (Chair of Sensor-based Geoinformatics, University of Freiburg, 2026; Guimarães et al., 2020; WaldAgil, University of Freiburg, 2026a; White et al., 2016), the present benchmark focuses on TLS so that the proximal geometric reference remains consistent across both sites (Béland et al., 2019; Calders et al., 2020; Newnham et al., 2015; Wilkes et al., 2017).

Instance segmentation is addressed by SegmentAnyTree, a sensor- and platform-agnostic pipeline for individual trees (Wielgosz et al., 2024), and by ForestMamba, a structure-aware model combining sparse Mamba state-space modelling with geometry-guided query initialization (Nguyen et al., 2026). Closely related automated-inventory and ground-based instance methods provide additional context for the benchmark design (Cherlet et al., 2026; Henrich et al., 2024; Puliti et al., 2023; Xiang et al., 2024, 2025). Species classification is addressed by DetailView, a multi-view DenseNet architecture for tree species prediction (Puliti et al., 2025).

For evaluation, field inventory species labels are mapped to the FOR-species20K taxonomy IDs used by DetailView. Inventory labels that cannot be mapped to the local label set are excluded from species accuracy. Manual instance references were delineated with CloudCompare using the terrestrial point clouds. Each reference package provides stable instance identifiers and species prediction dimensions that allow direct comparison across instance sources.

The two study plots differ in footprint size and effective point density per unit area. In the provided tree-point reference packages, Ettenheim covers $50\text{ m} \times 50\text{ m}$ with about 20.5 million tree points (about 8.2 k points per square meter). Mathislewald covers $25\text{ m} \times 25\text{ m}$ with about 12.9 million tree points (about 20.7 k points per square meter). This density and annotation feasibility consideration informed the choice of a larger Ettenheim footprint and a smaller Mathislewald footprint while keeping both sites representative for their local species composition.

4 Methodology

The Study sites and data section establishes that field inventory and terrestrial laser scanning (TLS) were already available for Ettenheim and Mathislewald. The methodological chain below

therefore starts from those source products and documents how the analysis footprint was narrowed, how reference instances were built, how segmentation models were compared to that reference, how inventory stems were linked to manually delineated trees and how species classification was evaluated on the resulting matched set. All geospatial processing used the projected coordinate reference system EPSG:32632.

4.1 Inventory-guided spatial narrowing of the TLS clouds

The delivered TLS scenes extend beyond the inventoried stems. To restrict subsequent work to the area that is jointly supported by inventory and proximal geometry, inventory stem positions were treated as a planar point set and enclosed with a concave hull. A fixed buffer of 10m was then applied around that hull, and the buffered polygon was used to clip the TLS point cloud.

At Ettenheim the concave hull was computed with Shapely’s concave-hull routine using a ratio of 0.05 and without holes. At Mathislewald an adaptive ratio schedule was used (0.2, 0.15, 0.25, 0.3, 0.1, 0.35, 0.4, 0.5, 1.0) until the hull perimeter contained at most 120 vertices. If no ratio satisfied that constraint the convex hull was used instead. In both cases the clip polygon was required to cover every valid inventory point. Clipping was performed in streaming fashion with `laspy` so that only points whose planimetric coordinates fell inside the buffered polygon were retained.

Bird’s-eye verification plots were produced for both sites by overlaying a thinned view of the source point cloud, the inventory stem locations, the inner concave hull outline and the buffered clip boundary (Figure 3). These maps confirmed that the retained cloud surrounds the inventoried stand with a controlled margin rather than truncating stems at the inventory extent.

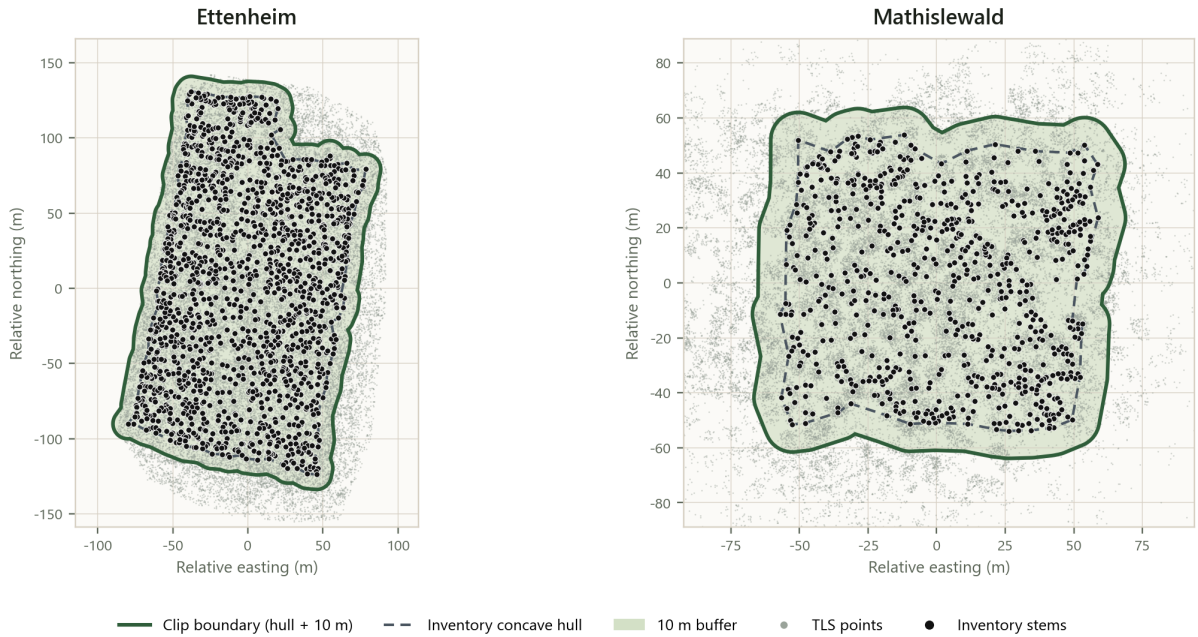


Figure 3: Inventory-guided TLS clipping for Ettenheim (left) and Mathislewald (right). Inventory stems, the inventory concave hull, the 10m buffer and the resulting clip boundary are shown over a thinned top view of the TLS cloud.

4.2 Subdivision into labelled subplots

The inventory-clipped clouds remain large relative to interactive three-dimensional editing. Each clipped scene was therefore partitioned into square subplots of $25\text{ m} \times 25\text{ m}$ (625 m^2). The subplot grid was centred on the clip-polygon centroid. Only cells that intersected the clip polygon and retained more than 1 m^2 of clipped area were kept. Subplots were numbered in a ring-clockwise order starting from the centre (`tile_001`, `tile_002`, `tile_003`, ...).

Within each subplot, preliminary ForestMamba instance labels already present in the source clouds were used to export one LAZ file per predicted tree, together with ground and residual catch-all layers. Trees with any point within 0.12 m of the outer clip boundary were excluded from the valid tree set so that incomplete edge crowns would not enter later reference construction. The resulting subplot layouts were mapped and labelled for both sites (Figure 4).

Manual reference construction and all subsequent benchmarks reported in this thesis focus on the subplots that were completed as CloudCompare projects: Ettenheim `tile_001`-`tile_004` (approximately $50\text{ m} \times 50\text{ m}$ of analysed footprint) and Mathislewald `tile_001` (approximately $25\text{ m} \times 25\text{ m}$). The remaining subplots of the clipped scenes were retained as an organised archive but were not manually annotated. Plot-scale TLS acquisition and interactive editing of dense temperate scenes remain labour-intensive even with modern scanners (Kükenbrink et al., 2022; Liang et al., 2016; Newnham et al., 2015; Wilkes et al., 2017).

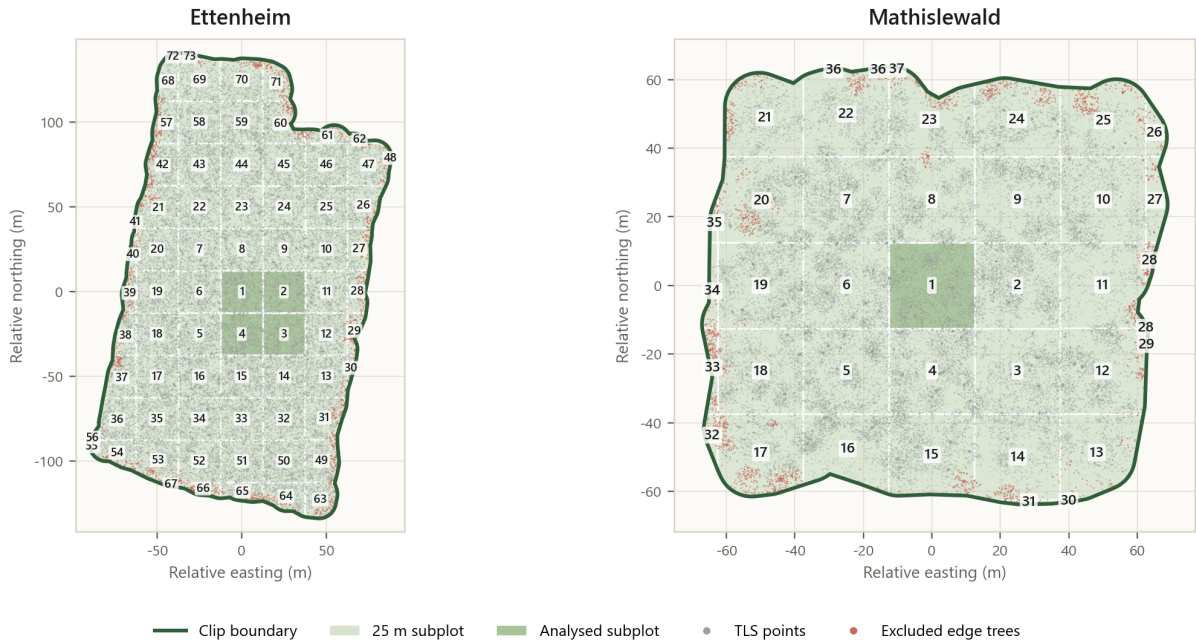


Figure 4: Twenty-five-metre subplot layouts for Ettenheim (left) and Mathislewald (right). Subplots are numbered in ring-clockwise order from the centre. Darker cells mark the manually analysed subplots, and coral points mark trees excluded for lying on the outer clip edge.

4.3 Manual instance delineation in CloudCompare

For each analysed subplot the exported LAZ content was imported into CloudCompare and saved as a modern CloudCompare project (`.bin`, CCB2). Starting from the preliminary ForestMamba

instances, layers were interactively split, merged and cleaned until each retained tree layer corresponded to a single physical stem and crown. Residual non-tree or ambiguous material was kept in dedicated catch-all layers (for example `ground_sem0`, `misc_inst-1` and related `misc` groups) rather than being forced into a tree identity.

After editing, project hierarchy was standardised so that each tree group contained one cloud, path suffixes were stripped from display names and cloud names were aligned with their parent groups. Tree layers followed a stable `tree_#####` naming pattern. Where a preliminary multi-tree blob was split into several physical trees, sibling layers were named `tree_#####_1`, `tree_#####_2`, ... so that each retained layer is one ground-truth tree (the numeric base is not a shared instance identity). Each completed project was spot-checked in the CloudCompare graphical interface by listing the layer tree and inspecting whether any retained tree layer still mixed multiple stems or left an obvious crown incomplete.

These manually refined CloudCompare layers constitute the geometric ground truth for instance-segmentation evaluation. Inventory matching, described later, is a separate step used for species evaluation and is not required to define the segmentation reference. Classical geometric reconstruction tools remain available for related structural workflows (Burt et al., 2019; Hackenberg et al., 2015; Lowe and Stepanas, 2021; Raunonen et al., 2013), but were not used as the geometric reference here.

4.4 Instance-segmentation evaluation protocol

Automatic instance predictions were compared with the manual CloudCompare reference under a shared point support. All points belonging to the analysed subplot projects (tree layers, catch-alls and ground) defined the evaluation set X_{eval} . The same XYZ locations were then extracted from the ForestMamba and SegmentAnyTree products so that every point carried a manual ground-truth instance identifier and a predicted instance identifier. Ground, catch-all and non-tree material received ground-truth identifier -1. Predicted SegmentAnyTree labels used in the final comparison were taken from absolute-coordinate partner products (`PredInstance_SAT`) rather than from earlier height-normalized transfers.

Matching followed the individual-tree evaluation style of ForestFormer3D (Xiang et al., 2025), consistent with related automated-inventory and TLS benchmarking protocols (Cherlet et al., 2026; Puliti et al., 2023; Wang et al., 2016; Xiang et al., 2024). Point-count Intersection-over-Union (IoU) was computed between each predicted instance and each ground-truth tree. Predicted instances with fewer than 100 points were discarded in the primary protocol (a common noise filter for tiny fragments). As a sensitivity check we repeated plot-level matching at $n_{\text{min}} \in \{0, 25, 50, 100, 200\}$. Remaining predictions were assigned to ground-truth trees by greedy one-to-one maximisation of IoU. A match was accepted only when IoU was at least $\tau = 0.5$. As a further sensitivity check we also solved the same IoU matrix with the Hungarian algorithm (global bipartite assignment maximising total IoU, still rejecting pairs below τ) and compared plot-level precision, recall, F1 and coverage at $\tau = 0.5$. The primary reported tables and figures retain greedy matching with $n_{\text{min}} = 100$. Label transfer onto X_{eval} used exact millimetre XYZ

hashing where possible and nearest-neighbour fill within 0.05 m otherwise.

From the matched assignments we computed precision, recall, F1-score and coverage (mean over ground-truth trees of the best IoU, without the τ gate), together with true-positive, false-positive and false-negative counts. Secondary diagnostics recorded over-segmentation and under-segmentation patterns using multi-overlap inspection at $\text{IoU} \geq 0.25$. The same matching was repeated for a threshold sweep $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ and for per-subplot strata. Plot-level F1 pools true positives, false positives and false negatives across the subplot. Separately, *per-tree hard F1* scores each ground-truth tree under the same greedy $\tau = 0.5$ assignment: when a tree is matched, hard F1 is the pairwise point-count F1 between that tree and its matched prediction. When it is unmatched, hard precision, recall and F1 are set to zero. *Macro hard F1* is the unweighted mean of those per-tree hard F1 scores over all N_{GT} reference trees, so missed trees pull the mean down rather than disappearing from the denominator. Soft per-tree F1 (best-overlap partner without the exclusive τ gate) was retained only as a secondary diagnostic. Per-tree hard F1 was then related to three structural covariates derived from each manual ground-truth tree cloud, so that later interpretation can separate detection failures from delineation failures. *Height* is the vertical extent $z_{99} - z_{05}$ of all points of that tree. The *whole-tree planimetric centroid* is the mean (x, y) of those same points (stem, crown and any attached woody material retained in the layer). It is not a breast-height stem position. *Nearest-neighbour distance* is the planimetric distance from that whole-tree centroid to the closest other ground-truth whole-tree centroid within the analysed footprint. As a density sensitivity check we also counted other ground-truth whole-tree centroids within a 5 m planimetric radius, censoring focal trees whose centroids lay within 5 m of the 25 m subplot boundary so that search discs were not truncated by missing off-tile reference trees. On that interior set the count did not clearly improve on nearest-neighbour distance for per-tree hard F1, so reported packing associations retain nearest-neighbour distance as the primary neighbourhood covariate. As a further packing check on the same interior focals we also computed each tree’s planimetric whole-tree convex-hull overlap fraction (share of the focal 2D hull area intersecting the union of other ground-truth hulls); that overlap measure likewise did not clearly improve on nearest-neighbour distance once residualised on spacing, so it is not promoted as a main packing covariate. *Whole-tree convex-hull volume* is the volume of the three-dimensional convex hull of the tree’s XYZ points (subsampled to at most 50 000 points when larger). It therefore encloses the entire retained tree object rather than a crown-only subset, and it is not derived from semantic crown segmentation. Breast-height layer centroids used later for inventory matching are a separate geometry and are not the positions underlying the nearest-neighbour covariate. Associations used two-sided Spearman rank correlations on per-tree scores. Trees within a subplot are spatially dependent, so N trees are not N independent experimental replicates (Ettenheim: four subplots; Mathisewald: one). We therefore treat that battery as exploratory: effect sizes (tercile means and ρ) are emphasised over binary significance tests, ρ and p are reported for all associations including those with no clear evidence under $p < 0.05$, and Benjamini–Hochberg false-discovery-rate q -values within each site provide a multiplicity-aware sensitivity summary. ForestFormer3D itself was not run as a third predictor. It provided only the evaluation template.

4.5 Visual co-registration of inventory and manual trees

Before formal inventory assignment, inventory stem positions were inspected against the manually segmented trees in plan view. A local ground surface was fitted to the manually labelled ground points of each subplot and a thin breast-height slice was extracted at 1.3 m above ground along the local surface normal, with a tolerance of ± 0.05 m. Inventory stems were drawn as circles whose radii reflected recorded diameters at breast height.

These overlays were used to check whether residual geospatial offset between inventory coordinates and the TLS scene would systematically prevent matching. Where a stem was visibly offset from its inventory marker but clearly represented the same tree on the basis of position, height and diameter context, the association was retained for later assignment rather than discarded as a geolocation conflict. At Mathislewald a translational shift of the inventory overlay was estimated from paired stem markers before final matching. At Ettenheim interactive overlay adjustment was available for the same visual check. Figure 5 shows the breast-height overlays used for this inspection at both sites.

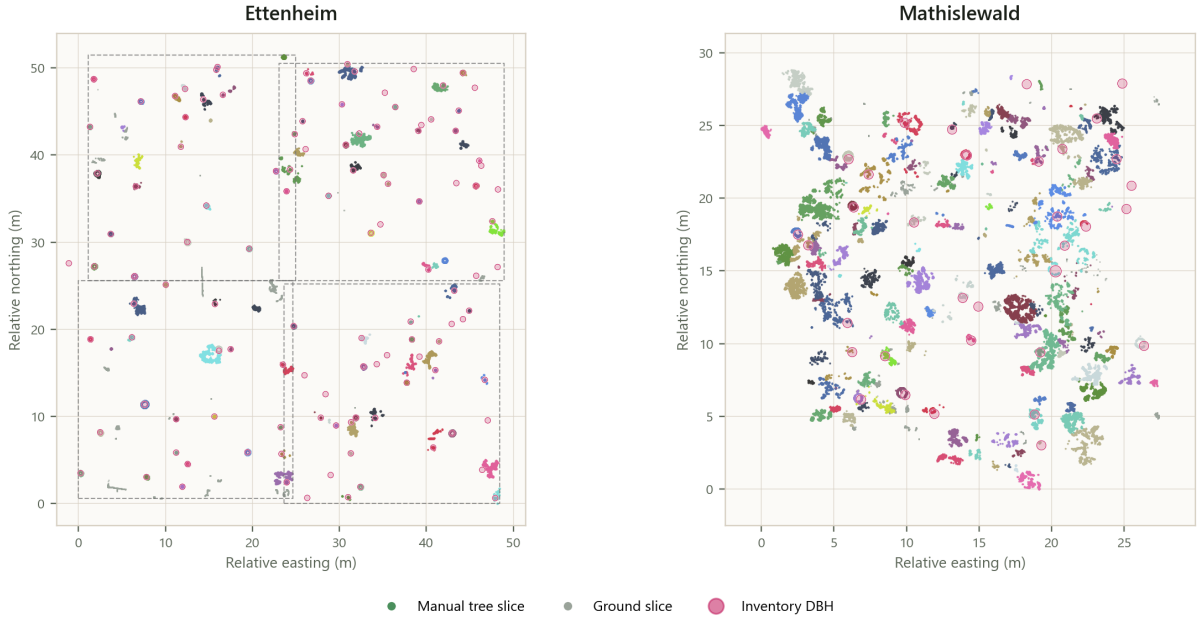


Figure 5: Breast-height plan views used to inspect co-registration of inventory and TLS at Ettenheim (left) and Mathislewald (right). Coloured points are manual tree layers in a $1.3 \text{ m} \pm 0.05 \text{ m}$ band along the local ground normal. Magenta rings mark inventory stem positions (radius proportional to recorded DBH, with a small minimum display size for readability). At Mathislewald the inventory circles incorporate the estimated translational shift.

4.6 Exclusive assignment of inventory stems to trees

Inventory stems and manual tree layers were then linked by an exclusive one-to-one rule on the same breast-height geometry. For each inventory stem and each manual layer the planimetric distance between stem centre and layer breast-height centroid was required to satisfy

$$\text{dist} \leq r_{\text{inv}} + r_{\text{layer}} + 0.75 \text{ m}, \quad (1)$$

where r_{inv} is the inventory radius at breast height and r_{layer} is the radius of the layer’s breast-height point envelope. Among geometrically eligible candidates, assignment was exclusive: each inventory stem and each layer could enter at most one pair.

When several layers lay within 0.75 m of the nearest eligible distance to the same inventory stem, ties were broken with an attribute cost based on the logarithmic diameter ratio and the relative height difference between inventory and layer. Incomplete stubs shorter than 3 m were strongly penalised when competing for a real inventory stem, and layers with more points were mildly preferred when other cues were similar. A small number of visually confirmed but geometrically awkward pairs were entered through an explicit override list after inspection, while layers known to be unsuitable for species evaluation were excluded from the matched set.

Unmatched inventory stems and unmatched manual layers were reviewed separately. Unmatched inventory positions were visualised as vertical marker poles in the three-dimensional scene to support inspection of possible missed trees, incomplete delineations or inventory points outside the annotated subplot. Unmatched manual layers were inspected for trees without an exclusive inventory partner (including small stems, stubs and edge trees) and for non-tree residuals. Only exclusive inventory-matched manual trees were retained for species classification. Species-classification performance is therefore conditional on successful inventory-to-instance registration (including visual overrides where used) and should not be interpreted as plot-wide classification accuracy on every manual or automatic tree. Stable instance identifiers were then written for those stems: at Ettenheim $\text{PredInstance} = \text{plot} \times 1000 + \text{tree}$, and at Mathislewald PredInstance followed the inventory tree identifier.

4.7 Species classification with DetailView

DetailView (Puliti et al., 2025) was applied to the inventory-matched packages (inference support is acknowledged separately). Inference used the European model region, took **PredInstance** as the tree identifier and returned point-wise species identifiers together with associated probabilities. Probabilities were averaged over 50 augmented forward passes per tree, following the DetailView practice used in the FOR-species20K benchmark setting. Restricting species evaluation to inventory-matched stems follows the broader observation that taxonomic products inherit both geometric delineation quality and the completeness of field labels (Fassnacht et al., 2016; Pu, 2021; Puliti et al., 2025).

To quantify how instance source affects taxonomic prediction on a fixed stem set, DetailView was run on four instance tracks that share the same inventory-matched stems: (i) the manual CloudCompare instances, (ii) SegmentAnyTree instances, (iii) ForestMamba instances and (iv) ForestMamba instances after an additional BluePoint refinement pass intended to reduce previously unresolved clusters. The returned point clouds preserved the original geometry and stored separate **species_id** and **species_prob** dimensions for each track. Trees missed by automatic segmentation outside that matched set do not enter the stem-level scores, so the ablation measures classification given instances rather than end-to-end species-map accuracy.

Local analysis then aggregated each track to the stem level by taking the mode of the non-nodata

species identifier inside every `PredInstance` (nodata code 255 ignored). Inventory species names were mapped to the FOR-species20K taxonomy used by DetailView (European beech $\rightarrow$ 10, silver fir $\rightarrow$ 0, Norway spruce $\rightarrow$ 14, Douglas fir $\rightarrow$ 25, European larch $\rightarrow$ 13). Predictions outside the locally present mapped classes were treated as open-set relative to the plot label set.

Stem-level comparisons were summarised with overall accuracy, balanced accuracy, macro-averaged precision, recall and F1-score and weighted F1-score. Uncertainty for overall accuracy on each site-level track was assessed with percentile bootstrap intervals (1000 resamples, fixed seed). Differences between the manual track and each automatic track on the same stems were summarised with paired percentile bootstrap intervals on the stem-wise accuracy gap ΔOA (and likewise for macro F1 where reported), resampling shared stem indices with replacement. Complementary summaries included row-normalized confusion matrices for every instance track, stem-level label agreement between tracks, per-class F1 by instance source, a conifer-versus-broadleaf collapse, overall accuracy by merged height strata (adjacent 2 m bins combined until each stratum has at least five stems, with Wilson 95% intervals), and overall-accuracy gaps that contrast the manual-instance track with each automatic-instance track on the same stems. Platform splits such as TLS versus mobile or unmanned laser scanning were not applicable because each plot uses a single proximal TLS modality.

4.8 Analysis products and intended inference

The pipeline yields three families of analysis products. First, subplot-wise and pooled instance-segmentation tables and figures quantify detection and delineation of ForestMamba and SegmentAnyTree against the manual CloudCompare reference, including behaviour across IoU thresholds and structural covariates. Second, exclusive match manifests linking inventory stems to trees document which stems enter the taxonomic evaluation and which inventory or layer residuals remain unmatched after geometric and visual review. Third, DetailView stem-level tables and figures quantify taxonomic agreement with inventory labels under manual and automatic instance sources and isolate the change attributable to switching the instance track. Inventory-matched stems were further joined to the height, whole-tree nearest-neighbour and whole-tree convex-hull-volume covariates of their matched manual layers so that species accuracy could be stratified by the same structural axes used for instance F1, again using exploratory two-sided Spearman tests with Benjamini–Hochberg q -values within site as a multiplicity-aware sensitivity summary.

These products are intended to answer, without circular use of inventory as geometric ground truth, (i) how well current instance-segmentation models recover manually verified trees in these temperate TLS scenes under this reference, (ii) how accurately DetailView recovers inventory taxonomy when instances are taken from the manual reference on matched stems and (iii) how much changing the instance source on those same stems changes stem-level species classification relative to the manual-instance baseline. Numerical outcomes are reported in the Results section.

5 Results

5.1 Observational overview of the prepared scenes

After inventory-guided concave-hull clipping and 25 m subplotting, the analysed footprints are the manually completed CloudCompare projects: Ettenheim subplots `tile_001`-`tile_004` and Mathislewald subplot `tile_001`. In plan view the inventoried stems sit inside the buffered clip, and the annotated subplots occupy the central portion of that clip (see Methodology Figures 3 and 4).

Qualitative instance colourings on the first subplot of each site show the same point support under three labelings: the manual CloudCompare reference, ForestMamba and SegmentAnyTree (Figures 6 and 7). Crowns and stems appear as discrete coloured objects in the manual panels. The automatic panels place predicted instances on the same geometry. At Mathislewald the manual panel resolves many small crowns as separate colours, whereas both automatic panels leave large parts of that crowded subplot without a matching predicted instance.

Ettenheim: tile 001 instance colours (subsamped)

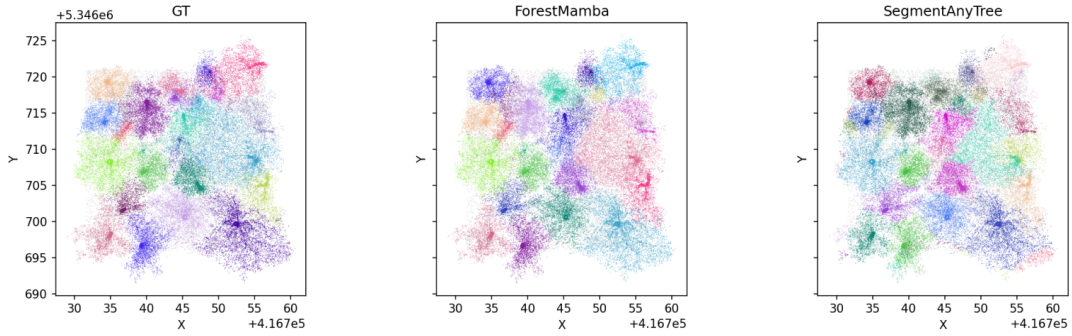


Figure 6: Ettenheim subplot `tile_001`: qualitative instance colouring for the manual reference, ForestMamba and SegmentAnyTree on a shared subsamped point support.

Mathislewald: tile 001 instance colours (subsamped)

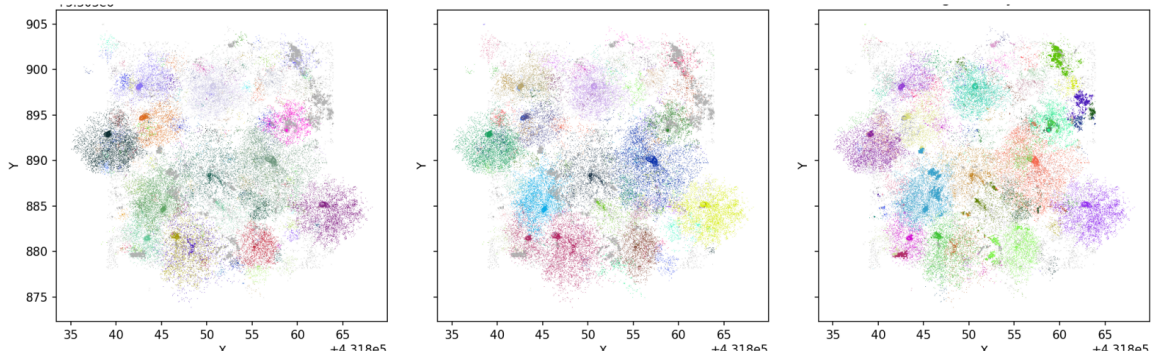


Figure 7: Mathislewald subplot `tile_001`: qualitative instance colouring for the manual reference, ForestMamba and SegmentAnyTree on a shared subsamped point support.

Breast-height plan views place inventory stem circles on the manually delineated layers at both

sites (Figure 8). On the $1.3\text{ m} \pm 0.05\text{ m}$ slice, inventory markers are distributed among the coloured manual trees. Some markers sit on or near a layer centroid, and others are offset from the nearest coloured stem while still corresponding to a visually identifiable tree in the surrounding cloud.

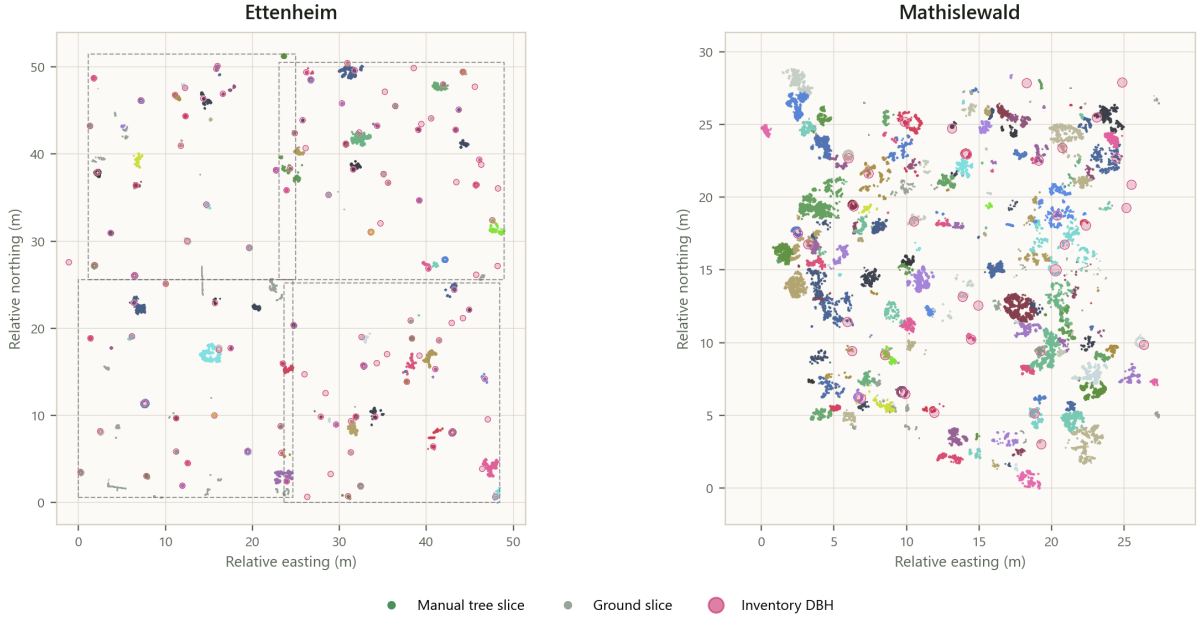


Figure 8: Breast-height overlay of inventory stem markers on manually delineated tree layers at Ettenheim (left) and Mathislewald (right) ($1.3\text{ m} \pm 0.05\text{ m}$ along the local ground normal). Magenta rings are inventory positions with radius proportional to DBH.

5.2 Manual reference scope

The segmentation reference comprises every retained manual tree layer in the analysed CloudCompare projects, with ground and catch-all material coded as background (`gt_id=-1`). Table 1 summarises the resulting evaluation supports.

Table 1: Manual instance-reference scope used for segmentation evaluation (N_{GT} population in Figure 9).

Site	Subplots	N_{GT}	Tree points	Total X_{eval} points
Ettenheim	001-004	128	22 714 670	26 997 057
Mathislewald	001	189	14 605 364	24 260 295

Per-subplot ground-truth tree counts at Ettenheim were 25, 41, 38 and 24 for `tile_001` to `tile_004`. At Mathislewald the single annotated subplot contributed 189 ground-truth trees (one per retained manual CloudCompare tree layer, including layers split from preliminary multi-tree blobs). These counts define the denominator for recall and related detection metrics below. They are distinct from the later inventory-matched stem counts used for species classification. Figure 9 summarises how those two evaluation populations relate on each site.

Evaluation populations: reference trees versus inventory-matched stems

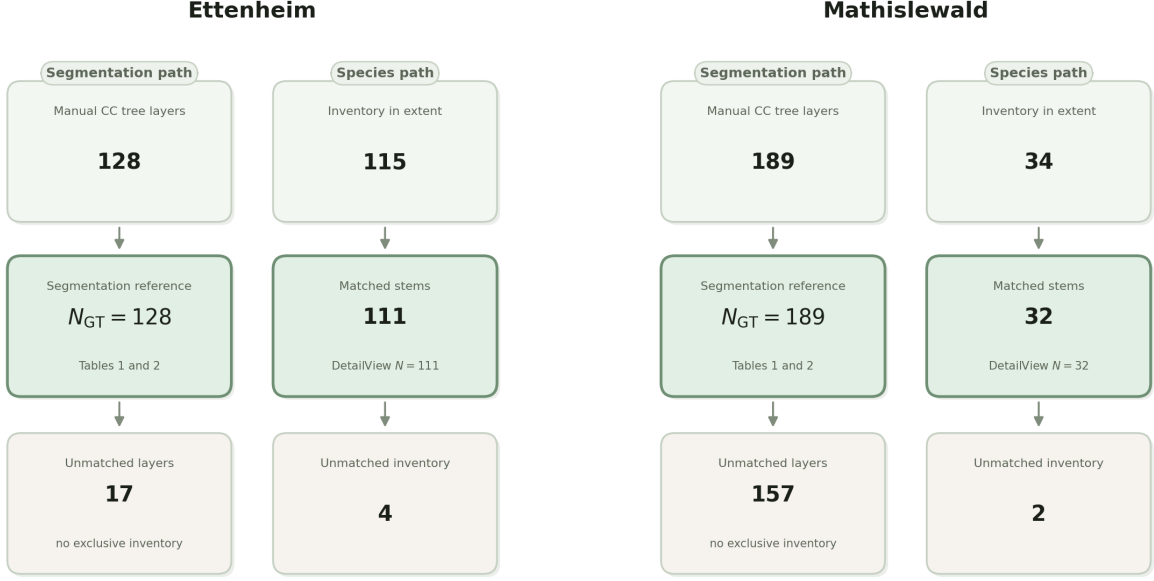


Figure 9: Evaluation populations on each site. The segmentation path uses every retained manual CloudCompare tree layer as N_{GT} (Tables 1 and 2). The species path retains only exclusive inventory-matched stems (Table 6), which are the DetailView evaluation set. Unmatched layers lack an exclusive inventory partner, and unmatched inventory stems lack an exclusive manual layer.

5.3 Overall performance

This subsection reports plot-level instance recovery, inventory match counts, and DetailView accuracy on the matched stems under the manual instance track (with a multi-track leaderboard for context). Structural covariates are deferred to the next subsection. Instance-source effects on species labels are deferred to the final Results subsection.

5.3.1 Plot-level instance segmentation

Matching used point-count IoU with greedy one-to-one assignment at $\tau = 0.5$ and a minimum of 100 predicted points per candidate instance. A Hungarian assignment on the same IoU matrices (still rejecting pairs below τ) reproduced the greedy plot-level precision, recall, F1 and coverage to numerical identity on both sites and both models at $\tau = 0.5$, so the primary tables retain the greedy protocol. Varying the minimum predicted-instance size over $n_{\min} \in \{0, 25, 50, 100, 200\}$ changed F1 by at most a few percentage points and did not alter model ranking or the Mathislewald SegmentAnyTree failure (F1 remained near 14–15% across thresholds). The default $n_{\min} = 100$ is therefore not driving the main conclusions. Figure 10 summarises precision, recall, F1-score and coverage for ForestMamba and SegmentAnyTree on both sites. Table 2 reports the corresponding values together with prediction counts and confusion counts.

At Ettenheim the per-subplot breakdown at $\tau = 0.5$ is summarised in Table 3 (Figure A.1). ForestMamba F1 ranged from 63.0% on `tile_004` to 72.7% on `tile_001`. SegmentAnyTree F1 ranged from 46.5% on `tile_003` to 65.4% on `tile_002`.

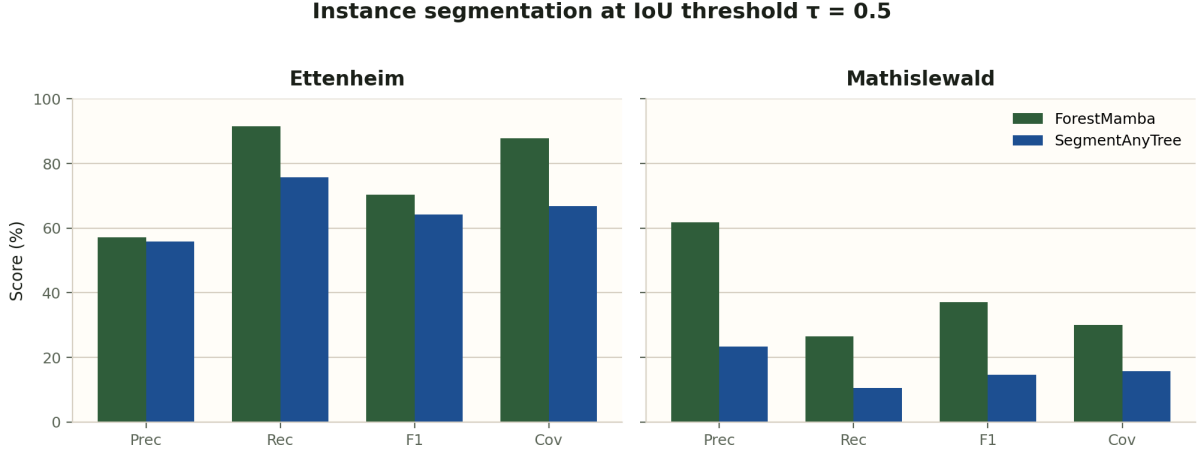


Figure 10: Plot-level instance-segmentation scores at IoU threshold $\tau = 0.5$ for ForestMamba and SegmentAnyTree on Ettenheim and Mathislewald.

Table 2: Instance-segmentation metrics at $\tau = 0.5$ on the full manual reference (N_{GT} ; Figure 9). Precision, recall, F1 and coverage are percentages.

Site	Method	Prec	Rec	F1	Cov	N_{pred}	TP	FP	FN
Ettenheim	ForestMamba	57.1	91.4	70.3	87.7	205	117	88	11
Ettenheim	SegmentAnyTree	55.7	75.8	64.2	66.9	174	97	77	31
Mathislewald	ForestMamba	61.7	26.5	37.0	29.9	81	50	31	139
Mathislewald	SegmentAnyTree	23.3	10.6	14.5	15.7	86	20	66	169

Table 3: Ettenheim per-subplot instance metrics at $\tau = 0.5$.

Subplot	N_{GT}	FM F1	FM Cov	SAT F1	SAT Cov	FM TP/FP/FN	SAT TP/FP/FN
001	25	72.7	94.3	59.5	76.8	24/17/1	22/27/3
002	41	71.8	88.4	65.4	71.9	37/25/4	34/29/7
003	38	69.5	80.3	46.5	52.9	33/24/5	23/38/15
004	24	63.0	91.4	56.3	76.1	23/26/1	20/27/4

Repeating the match over $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$ produced the curves in Figures A.2 and A.3. At Ettenheim, ForestMamba F1 moved from 70.9% at $\tau = 0.3$ to 70.3% at $\tau = 0.5$ and 66.1% at $\tau = 0.7$. SegmentAnyTree F1 moved from 68.9% to 64.2% to 50.3% over the same thresholds. At Mathislewald, ForestMamba F1 moved from 46.7% to 37.0% to 23.7%. SegmentAnyTree F1 moved from 26.2% to 14.5% to 7.3%.

Per-ground-truth maximum IoU distributions (Figures A.4 and A.5), error-taxonomy bars (Figures A.6 and A.7) and spatial TP/FN/FP centroid maps (Figures A.8 and A.9) are collected with the τ -sweeps in Appendix A. Error-taxonomy counts at $\tau = 0.5$ are summarised in Table 4. At Mathislewald the appendix plan views are dominated by false-negative ground-truth centroids for both models, with comparatively few true positives. ForestMamba recovers more matched trees than SegmentAnyTree, but neither leaves a spatially localised miss pattern confined to plot edges.

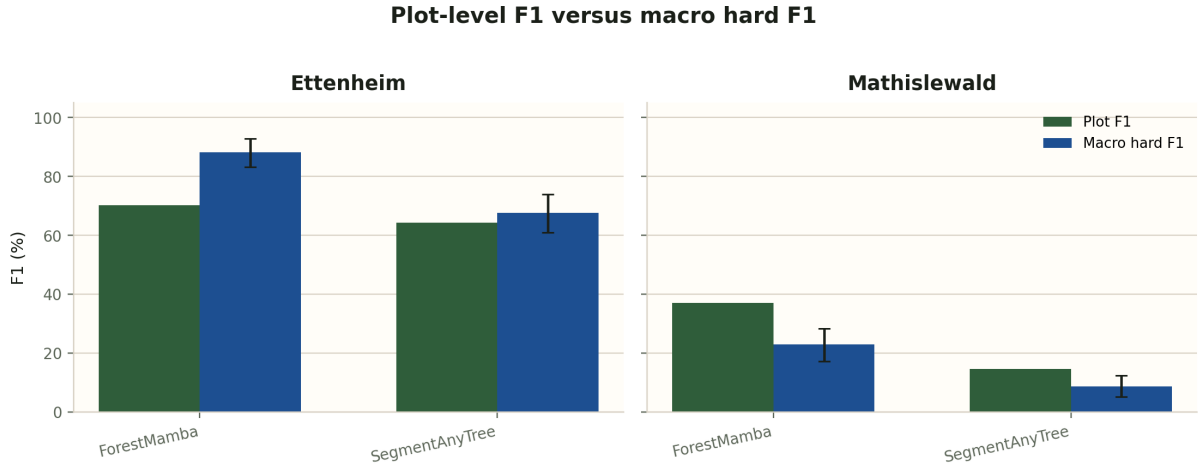
Table 4: Secondary error-taxonomy counts at $\tau = 0.5$.

Site	Method	Over-seg.	Under-seg.	Missed	Leakage-flagged TP
Ettenheim	ForestMamba	1	4	11	10
Ettenheim	SegmentAnyTree	1	4	31	42
Mathislewald	ForestMamba	0	6	139	18
Mathislewald	SegmentAnyTree	1	1	169	12

Macro hard F1 (mean of per-tree hard F1 scores) is reported beside plot-level F1 in Table 5 and Figure 11. At Ettenheim, ForestMamba macro hard F1 was 88.3% (95% CI 83.2-92.9) against a plot F1 of 70.3%. SegmentAnyTree macro hard F1 was 67.5% (60.9-74.0) against a plot F1 of 64.2%. At Mathislewald, ForestMamba macro hard F1 was 22.8% (95% CI 17.3-28.4) against a plot F1 of 37.0%. SegmentAnyTree macro hard F1 was 8.7% (5.1-12.3) against a plot F1 of 14.5%.

Table 5: Plot-level F1 versus macro hard F1 at $\tau = 0.5$.

Site	Method	Plot F1	Macro hard F1	95% CI
Ettenheim	ForestMamba	70.3	88.3	83.2-92.9
Ettenheim	SegmentAnyTree	64.2	67.5	60.9-74.0
Mathislewald	ForestMamba	37.0	22.8	17.3-28.4
Mathislewald	SegmentAnyTree	14.5	8.7	5.1-12.3

**Figure 11:** Plot-level F1 versus macro hard F1 for ForestMamba and SegmentAnyTree at Ettenheim and Mathislewald ($\tau = 0.5$). Error bars are bootstrap 95% intervals on macro hard F1.

5.3.2 Assignment of inventory stems to trees

Exclusive geometric matching on the breast-height slice linked inventory stems to manual layers under the rule $\text{dist} \leq r_{\text{inv}} + r_{\text{layer}} + 0.75 \text{ m}$, with attribute-based tie-breaking and a small override list as described in the Methodology. Figure 12 and Table 6 summarise the assignment outcomes. Figure 9 places those counts beside the segmentation N_{GT} population.

Exclusive matching retained 111 Ettenheim and 32 Mathislewald inventory stems (Table 6).

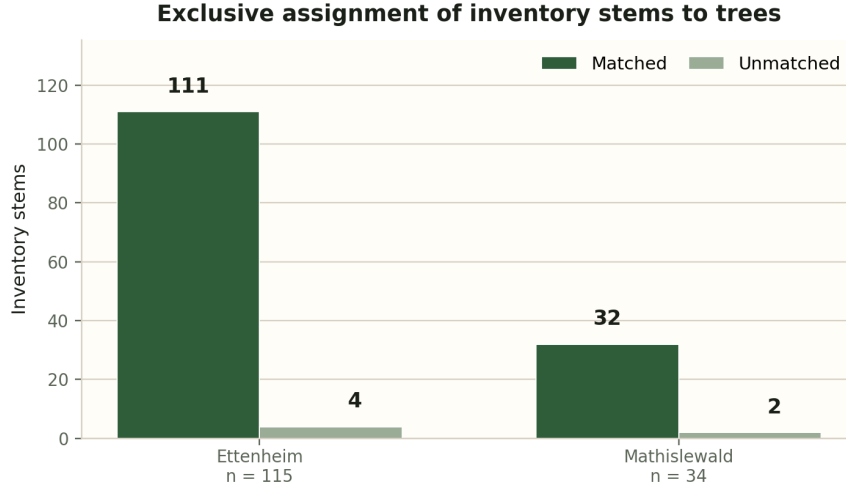


Figure 12: Counts of inventory stems that entered an exclusive match versus inventory stems that remained unmatched after geometric and visual review.

Table 6: Exclusive assignment counts linking inventory stems to trees on the analysed subplots (species / DetailView population in Figure 9).

Site	Inventory in extent	Matched	Unmatched inv.	Manual layers	Unmatched layers
Ettenheim	115	111	4	128	17
Mathislewald	34	32	2	189	157

Unmatched inventory positions were reviewed with vertical marker poles in the three-dimensional scene. Unmatched manual layers at Mathislewald (157 of 189) are mostly trees without an exclusive inventory partner (including small stems, stubs and other non-inventory residuals) relative to the short field list of 34 stems and the 32 exclusive matches.

5.3.3 Species classification with DetailView

DetailView was evaluated on the inventory-matched stems under four instance tracks: manual, SegmentAnyTree, ForestMamba and ForestMamba+BluePoint. Stem-level labels used the mode of non-nodata species identifiers inside each `PredInstance`. Figure 13 and Table 7 report overall accuracy and macro-averaged F1 for each site and track. Bootstrap percentile intervals for overall accuracy on each site-level row are given where available (1000 resamples).

5.3.4 DetailView on manual instances

On the manual-instance track, overall accuracy was 46.8% at Ettenheim ($N = 111$; bootstrap 95% CI 36.9-55.9) and 62.5% at Mathislewald ($N = 32$; 46.8-78.1). Macro F1 was 36.1% and 68.7% on the same tracks. Class supports on the matched sets were: Ettenheim European beech 79, silver fir 17, Douglas fir 13, European larch 1, Norway spruce 1; Mathislewald Norway spruce 21, European beech 7, silver fir 4.

Row-normalized confusion matrices for the manual track are shown in Figures 14 and 15. At Ettenheim the strongest diagonal entry was Douglas fir (92% of true stems predicted as Douglas

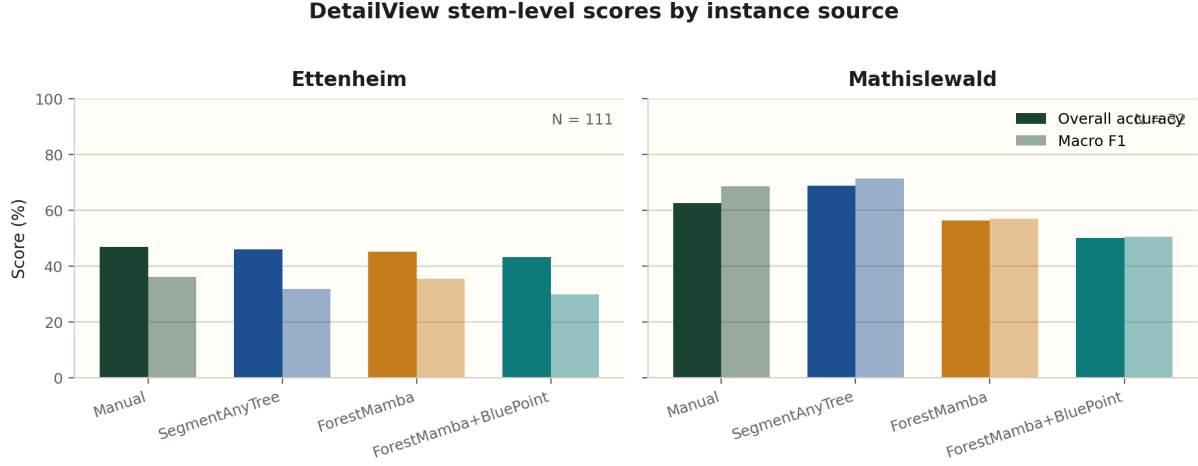


Figure 13: DetailView overall accuracy and macro F1 by instance source for Ettenheim ($N = 111$) and Mathislewald ($N = 32$).

Table 7: DetailView stem-level leaderboard. Overall accuracy (OA) and macro metrics are percentages. CI denotes the bootstrap 95% interval for OA.

Site	Instance source	N	OA	OA 95% CI	Macro P	Macro R	Macro F1
Ettenheim	Manual	111	46.8	36.9-55.9	49.6	35.1	36.1
Ettenheim	SegmentAnyTree	111	45.9	36.9-55.0	44.0	33.9	31.8
Ettenheim	ForestMamba	111	45.0	35.1-55.0	49.3	35.8	35.4
Ettenheim	ForestMamba+BluePoint	111	43.2	34.2-51.4	43.7	31.6	29.8
Mathislewald	Manual	32	62.5	46.8-78.1	86.7	57.9	68.7
Mathislewald	SegmentAnyTree	32	68.8	50.0-84.4	86.9	61.1	71.4
Mathislewald	ForestMamba	32	56.3	37.5-71.9	96.1	45.2	56.9
Mathislewald	ForestMamba+BluePoint	32	50.0	34.4-65.6	67.8	42.1	50.5
Both (pooled)	Manual	143	50.3	-	63.1	48.4	49.5
Both (pooled)	SegmentAnyTree	143	51.0	-	57.4	49.2	46.5
Both (pooled)	ForestMamba	143	47.6	-	61.8	49.4	47.8
Both (pooled)	ForestMamba+BluePoint	143	44.8	-	55.7	44.3	42.6

fir). European beech was 42% on the diagonal. Silver fir was split mainly between correct silver fir (41%) and Douglas fir predictions (35%). The single European larch stem and the single Norway spruce stem were each predicted entirely as Douglas fir. At Mathislewald, Norway spruce (67%), European beech (57%) and silver fir (50%) retained the largest correct fractions on the diagonal, with residual mass on neighbouring conifer and broadleaf labels.

5.4 Structural relationships

Per-tree hard F1 and DetailView correctness are related here to tree height ($z_{99} - z_{05}$), planimetric nearest-neighbour distance among whole-tree centroids, and whole-tree convex-hull volume.

Per-tree hard F1 was related to the three structural covariates defined in the Methodology: height ($z_{99} - z_{05}$), planimetric nearest-neighbour distance between whole-tree ground-truth centroids and whole-tree convex-hull volume (Figures 16, 17 and 18; Table 8). A fixed-radius neighbour count and a planimetric whole-tree hull-overlap fraction were also examined on edge-censored

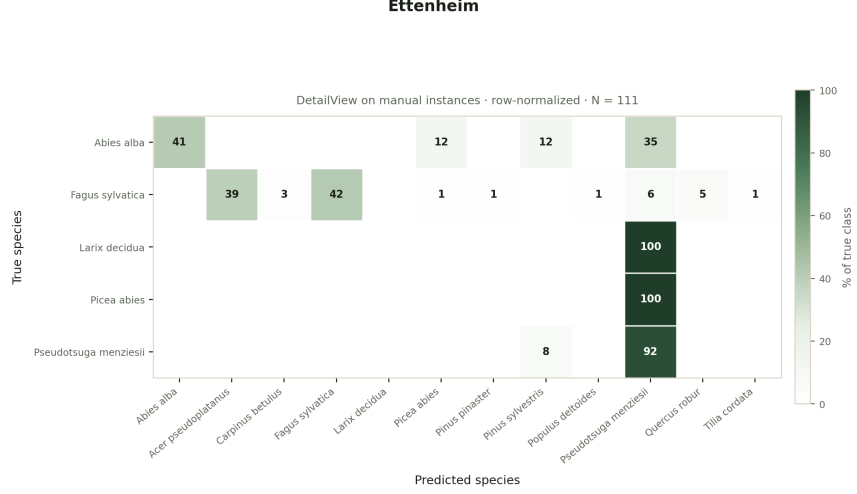


Figure 14: Ettenheim row-normalized DetailView confusion matrix on manual instances ($N = 111$).

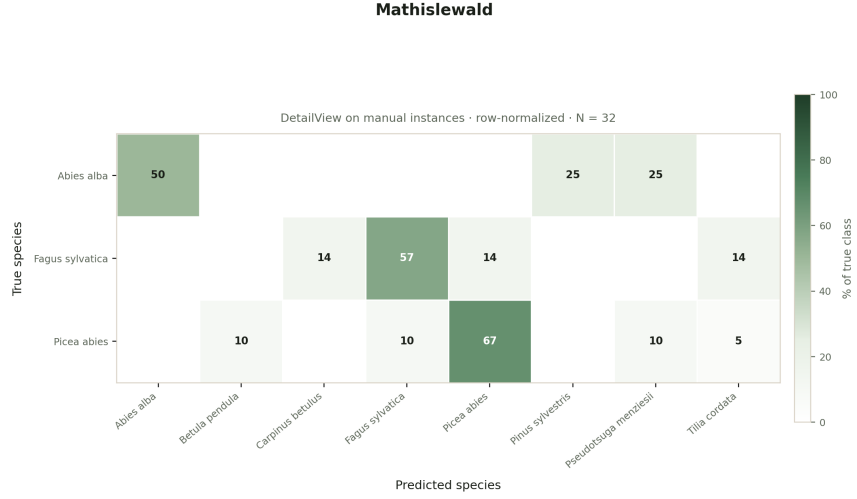


Figure 15: Mathislewald row-normalized DetailView confusion matrix on manual instances ($N = 32$).

interior trees as denser packing descriptors, but neither clearly improved on nearest-neighbour distance once residualised on that spacing, so packing associations below retain nearest-neighbour distance. Each point is one manual ground-truth tree. Black diamonds mark tercile means. Spearman rank correlations are annotated per panel. These Spearman tests are exploratory: trees within subplots are spatially dependent, effect sizes (ρ and tercile means) are emphasised over binary significance, associations with no clear evidence under two-sided uncorrected $p < 0.05$ are still quoted with their p -values, and Benjamini–Hochberg q -values within site are reported in the correlation tables as a multiplicity-aware sensitivity summary.

At Ettenheim ($N = 128$), hard F1 increased from short to tall trees for both models (ForestMamba tercile means 79.2% / 89.9% / 95.8%; SegmentAnyTree 51.5% / 64.1% / 86.9%). The Spearman association with height showed no clear evidence for ForestMamba ($\rho = 0.04$, $p = 0.68$) and met $p < 0.05$ for SegmentAnyTree ($\rho = 0.19$, $p = 0.032$). Nearest-neighbour distance among whole-tree centroids showed a clearer packing signal: crowded trees (low NN) had mean hard F1 of 75.0% (ForestMamba) and 41.9% (SegmentAnyTree), versus 97.5% and 87.2% in the most

open tercile. Spearman ρ was 0.28 ($p = 0.0016$) for ForestMamba and 0.41 ($p < 0.001$) for SegmentAnyTree. Partial correlations residualising on $\log(1 + n_{\text{points}})$ remained positive (0.25 and 0.35). Whole-tree convex-hull volume tracked the same direction as size (ForestMamba terciles 74.5% / 93.0% / 97.5%; SegmentAnyTree 46.6% / 74.2% / 82.0%), but neither correlation provided clear evidence under $p < 0.05$ (ForestMamba $\rho = 0.08$, $p = 0.36$; SegmentAnyTree $\rho = 0.17$, $p = 0.052$).

At Mathislewald ($N = 189$), height and whole-tree convex-hull volume again separated success more strongly than nearest-neighbour spacing, but absolute hard F1 was much lower than at Ettenheim because most small reference trees were missed. ForestMamba hard F1 rose from 3.6% in the lowest height tercile to 53.1% in the highest ($\rho = 0.57$, $p < 0.001$). SegmentAnyTree rose from 0.0% to 21.2% ($\rho = 0.42$, $p < 0.001$). Hull-volume terciles were 0.0% / 11.2% / 57.3% for ForestMamba ($\rho = 0.64$, $p < 0.001$) and 0.0% / 2.3% / 23.8% for SegmentAnyTree ($\rho = 0.44$, $p < 0.001$). Nearest-neighbour distance met $p < 0.05$ for ForestMamba ($\rho = 0.31$, $p < 0.001$; open-tercile mean 41.3% versus 13.5% crowded) but showed no clear evidence for SegmentAnyTree ($\rho = 0.07$, $p = 0.35$; open 10.4% versus crowded 7.1%).

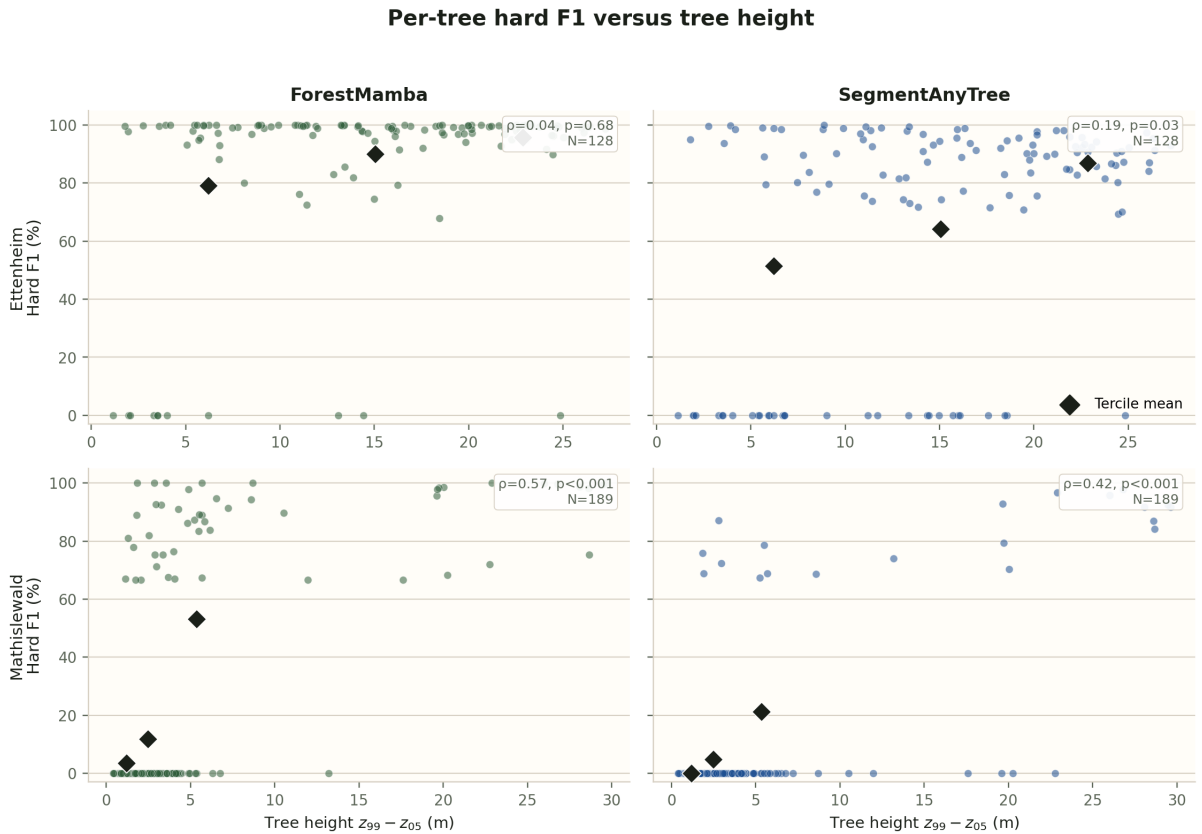


Figure 16: Per-tree hard F1 against tree height ($z_{99} - z_{05}$) for ForestMamba and SegmentAnyTree at Ettenheim (top) and Mathislewald (bottom). Diamonds are tercile means.

Overall accuracy by merged height strata (minimum five stems per stratum) and by lifeform (broadleaf versus conifer) is summarised in Figures 19 and 20. On the manual track, Ettenheim broadleaf overall accuracy was 41.8% ($N = 79$) and conifer overall accuracy was 59.4% ($N = 32$). Mathislewald broadleaf overall accuracy was 57.1% ($N = 7$) and conifer overall accuracy was

Per-tree hard F1 versus planimetric nearest-neighbour distance

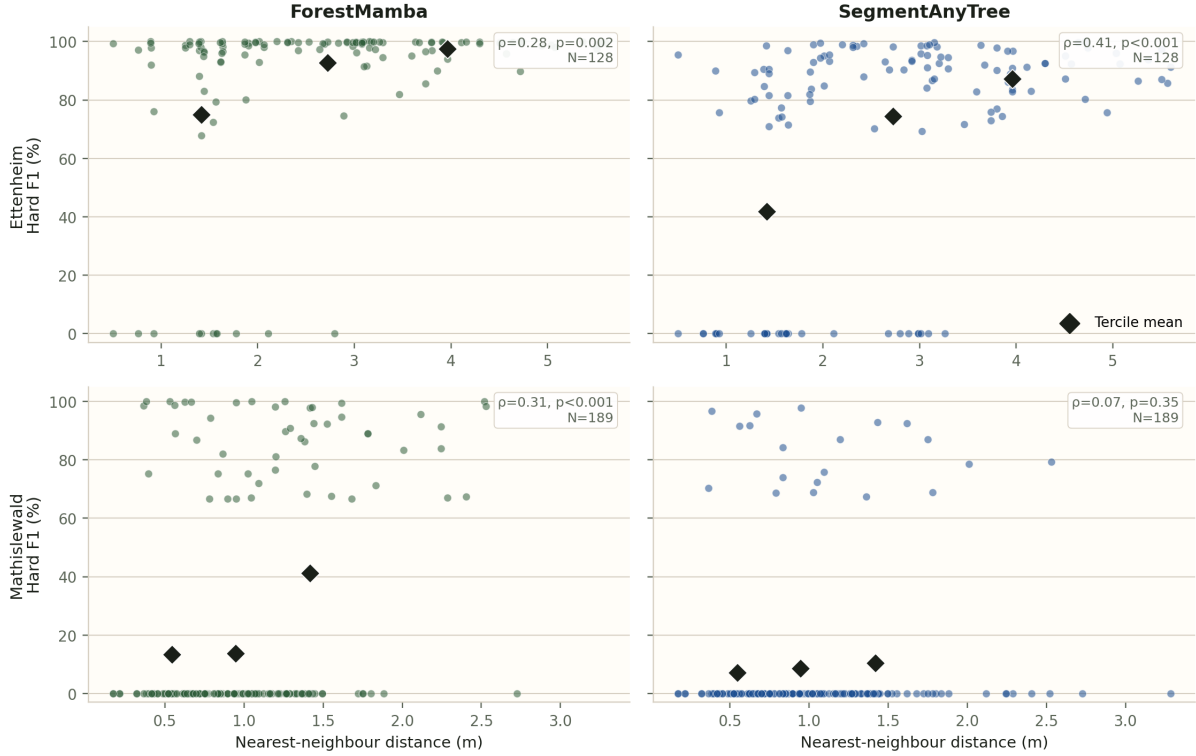


Figure 17: Per-tree hard F1 against planimetric nearest-neighbour distance between whole-tree centroids for ForestMamba and SegmentAnyTree at Ettenheim (top) and Mathislewald (bottom). Diamonds are tercile means.

Table 8: Mean per-tree hard F1 (%) by covariate tercile (low / mid / high). Tercile cuts are site-specific.

Site	Covariate	ForestMamba	SegmentAnyTree
Ettenheim	Height	79.2 / 89.9 / 95.8	51.5 / 64.1 / 86.9
Ettenheim	Nearest neighbour	75.0 / 92.8 / 97.5	41.9 / 74.5 / 87.2
Ettenheim	Whole-tree hull volume	74.5 / 93.0 / 97.5	46.6 / 74.2 / 82.0
Mathislewald	Height	3.6 / 11.8 / 53.1	0.0 / 4.8 / 21.2
Mathislewald	Nearest neighbour	13.5 / 13.7 / 41.3	7.1 / 8.6 / 10.4
Mathislewald	Whole-tree hull volume	0.0 / 11.2 / 57.3	0.0 / 2.3 / 23.8

64.0% ($N = 25$). Restricting to stems taller than 5 m gave manual-track overall accuracies of 46.3% at Ettenheim ($N = 108$) and 68.2% at Mathislewald ($N = 22$). Spearman association between stem height and manual-track correctness showed no clear evidence at either site under $p < 0.05$ ($\rho = -0.04$, $p = 0.66$ at Ettenheim; $\rho = 0.21$, $p = 0.25$ at Mathislewald).

The same inventory-matched stems were joined to the manual-cloud whole-tree nearest-neighbour distance and whole-tree convex-hull volume of their matched geometric layers (Figures 21 and 22; Table 9; point-level strip plots in Figures A.10 and A.11). At Ettenheim, manual-track overall accuracy was 42.1% in the most crowded NN tercile, 41.7% in the mid tercile and 56.8% in the most open tercile ($N = 38/36/37$). The Spearman link between NN distance and correctness was positive but showed no clear evidence under $p < 0.05$ ($\rho = 0.12$, $p = 0.21$). Automatic

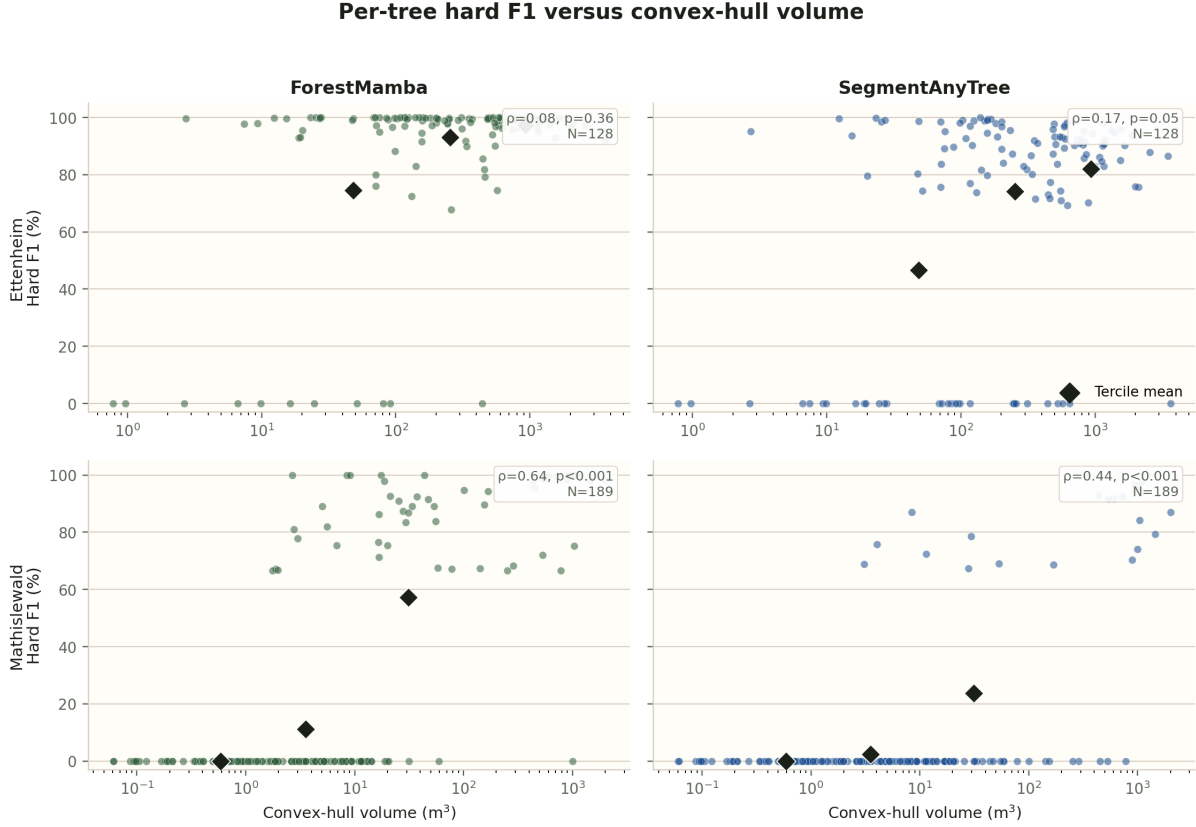


Figure 18: Per-tree hard F1 against whole-tree convex-hull volume ($\log x$) for ForestMamba and SegmentAnyTree at Ettenheim (top) and Mathisewald (bottom). Diamonds are tercile means.

tracks likewise showed no clear evidence (SegmentAnyTree $\rho = 0.14$, $p = 0.16$; ForestMamba $\rho = 0.14$, $p = 0.15$; ForestMamba+BluePoint $\rho = 0.13$, $p = 0.17$). Whole-tree convex-hull volume showed the opposite gradient: manual OA fell from 64.9% (low volume) through 45.9% (mid) to 29.7% (high volume), with Spearman $\rho = -0.28$ ($p = 0.0030$). ForestMamba ($\rho = -0.27$, $p = 0.0048$) and ForestMamba+BluePoint ($\rho = -0.20$, $p = 0.036$) reproduced the same decline. SegmentAnyTree showed no clear evidence under $p < 0.05$ ($\rho = -0.12$, $p = 0.21$).

At Mathisewald ($N = 32$), manual OA was 36.4% in the crowded NN tercile, 81.8% mid and 70.0% open ($\rho = 0.24$, $p = 0.18$). Automatic tracks also showed no clear evidence under $p < 0.05$ (SegmentAnyTree $\rho = 0.18$, $p = 0.32$; ForestMamba $\rho = -0.017$, $p = 0.93$; ForestMamba+BluePoint $\rho = -0.091$, $p = 0.62$). Hull-volume terciles were 45.5% / 80.0% / 63.6% with no clear rank association under $p < 0.05$ (manual $\rho = 0.17$, $p = 0.36$; SegmentAnyTree $\rho = -0.015$, $p = 0.94$; ForestMamba $\rho = -0.048$, $p = 0.80$; ForestMamba+BluePoint $\rho = -0.14$, $p = 0.46$). With about ten stems per bin, these Mathisewald strata are directional only.

5.5 Segmentation-to-species effects

Holding the inventory-matched stem set fixed and changing only the instance source isolates the taxonomic effect of automatic delineation on those stems. Missed trees outside the matched set are not scored. Overall-accuracy contrasts relative to the manual track are shown in Figure 23 and Table 10. At Ettenheim, overall accuracy on SegmentAnyTree instances was 0.9 percentage points

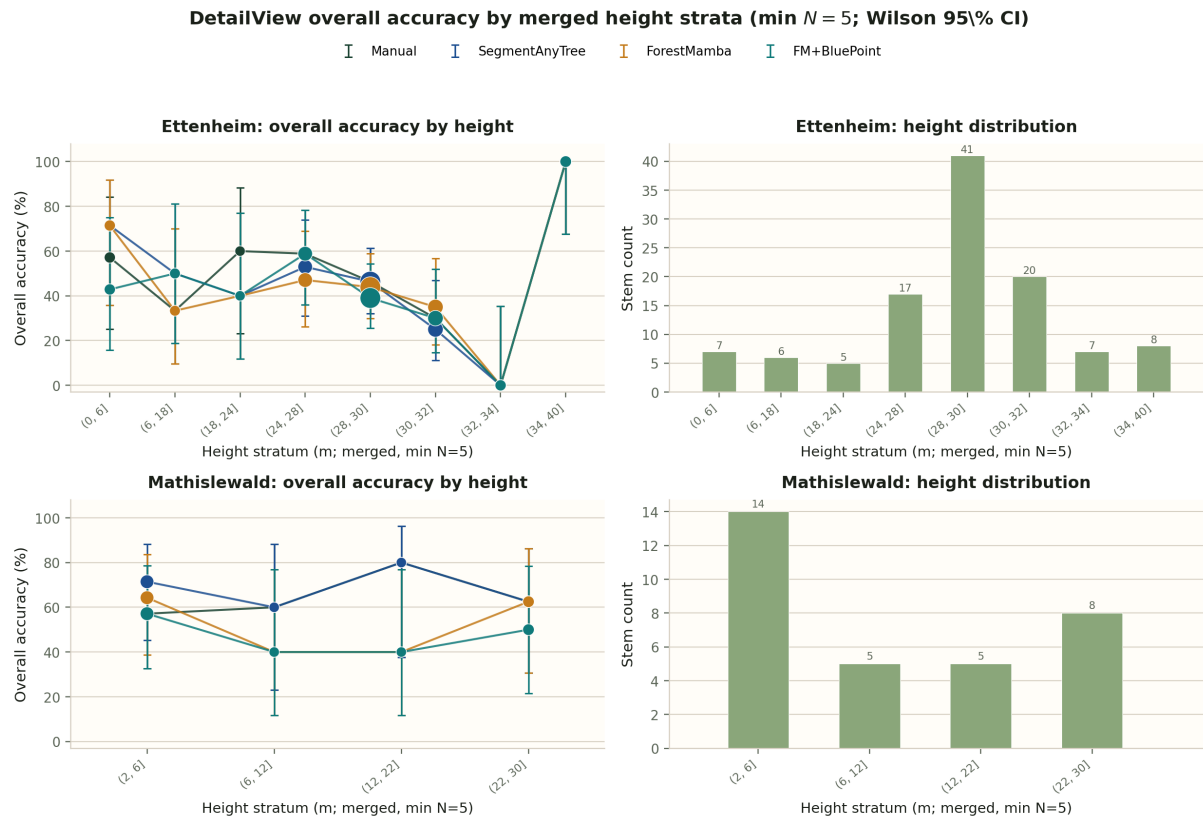


Figure 19: DetailView overall accuracy by merged height strata for each site and instance source (adjacent 2 m bins combined until $N \geq 5$; Wilson 95% intervals; marker size scales with N).

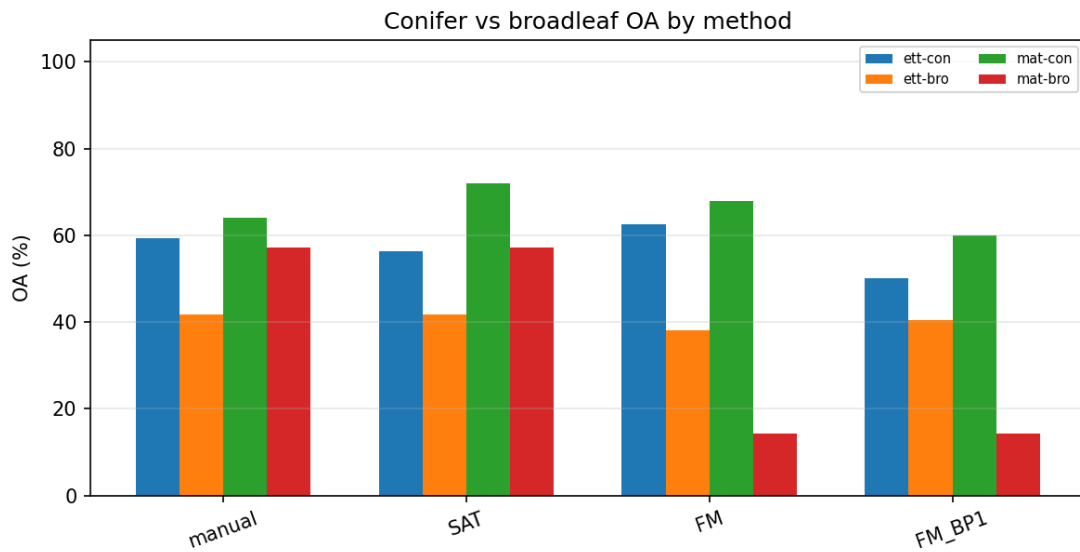


Figure 20: DetailView overall accuracy for broadleaf and conifer stems by site and instance source.

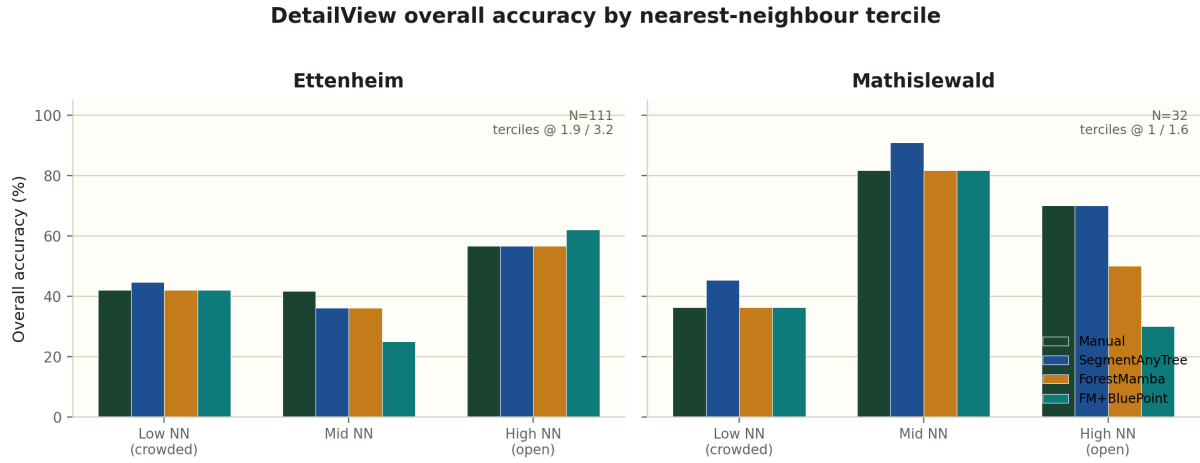


Figure 21: DetailView overall accuracy by nearest-neighbour tercile (site-specific cuts) for each instance source. Low NN denotes crowded trees (short distance between whole-tree centroids). High NN denotes more open neighbourhoods.

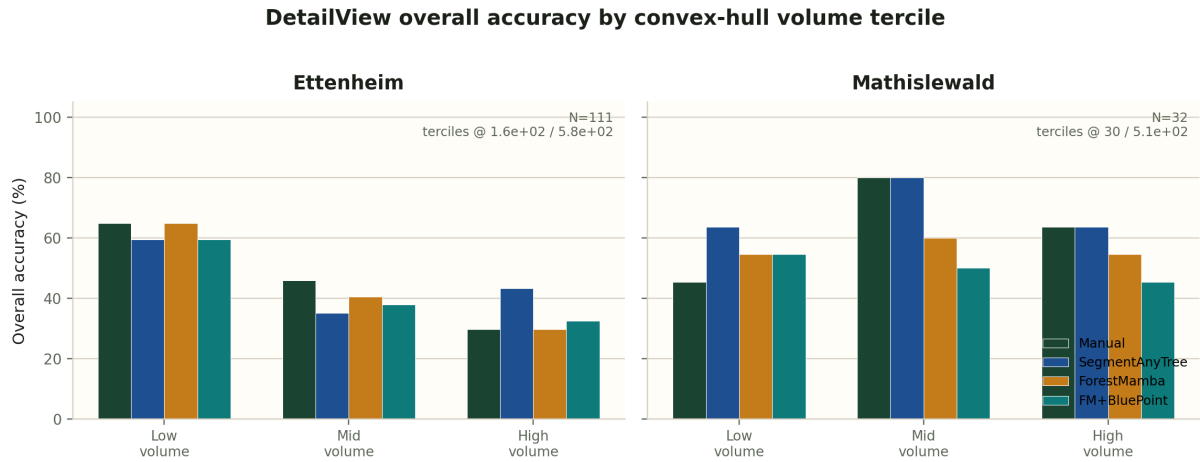


Figure 22: DetailView overall accuracy by whole-tree convex-hull volume tercile for each site and instance source.

Table 9: Manual-track DetailView overall accuracy (%) by structural tercile on inventory-matched stems.

Site	Covariate	Low / crowded	Mid	High / open
Ettenheim	Nearest neighbour	42.1	41.7	56.8
Ettenheim	Whole-tree hull volume	64.9	45.9	29.7
Mathislewald	Nearest neighbour	36.4	81.8	70.0
Mathislewald	Whole-tree hull volume	45.5	80.0	63.6

below the manual track (paired bootstrap 95% CI for ΔOA : -7.2 to $+9.0$). ForestMamba was 1.8 percentage points below (CI -3.6 to $+8.1$) and ForestMamba+BluePoint was 3.6 percentage points below (CI -3.6 to $+10.8$). The ForestMamba+BluePoint minus ForestMamba difference in overall accuracy was -1.8 percentage points (CI -9.0 to $+5.4$). All Ettenheim paired intervals include zero. At Mathislewald, overall accuracy on SegmentAnyTree instances was 6.3 percentage points above the manual track (CI -15.6 to 0.0). ForestMamba was 6.3 percentage points below (CI -6.3 to $+18.8$) and ForestMamba+BluePoint was 12.5 percentage points below (CI -3.1 to $+31.3$). The ForestMamba+BluePoint minus ForestMamba difference was -6.3 percentage points (CI -18.8 to $+3.1$). With $N = 32$, these Mathislewald intervals are wide. None of the positive manual-minus-automatic gaps exclude zero, while the SegmentAnyTree advantage versus manual reaches a CI upper bound of zero.

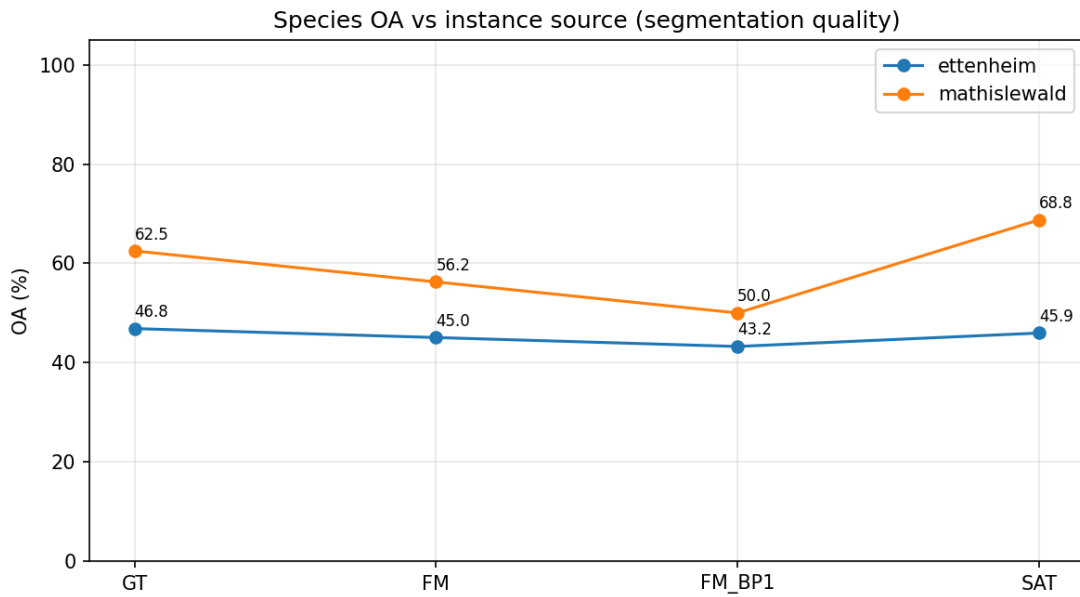


Figure 23: DetailView overall accuracy as a function of instance source for Ettenheim and Mathislewald.

Table 10: DetailView overall-accuracy differences ΔOA (percentage points; manual minus automatic, except the last row) with paired bootstrap 95% intervals on the shared stems.

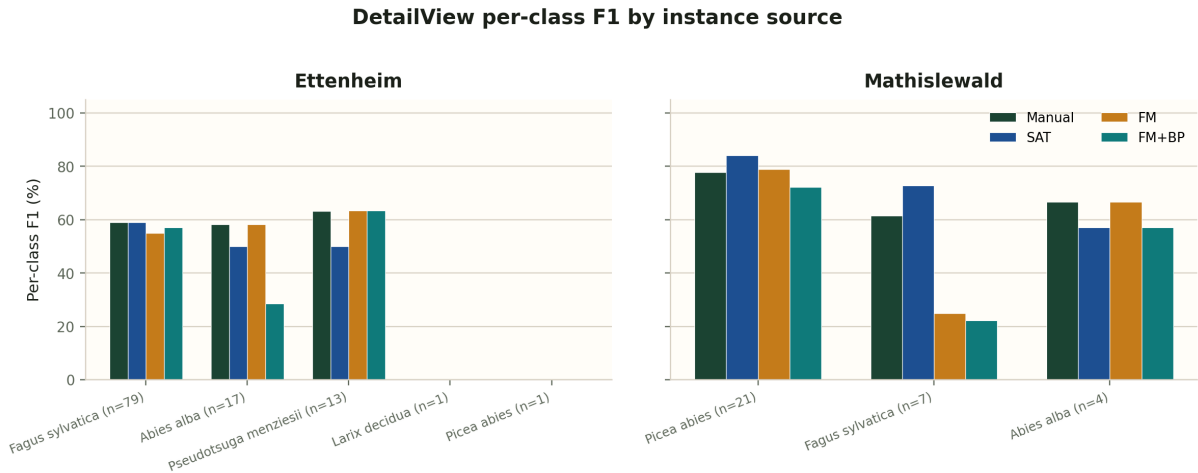
Contrast	Ettenheim	Mathislewald
Manual – SegmentAnyTree	+0.9 [−7.2, +9.0]	−6.3 [−15.6, 0.0]
Manual – ForestMamba	+1.8 [−3.6, +8.1]	+6.3 [−6.3, +18.8]
Manual – ForestMamba+BluePoint	+3.6 [−3.6, +10.8]	+12.5 [−3.1, +31.3]
ForestMamba+BluePoint – ForestMamba	−1.8 [−9.0, +5.4]	−6.3 [−18.8, +3.1]

Stem-level agreement between DetailView labels from different instance tracks is reported in Table 11 (Figure A.12). At Ettenheim, manual versus ForestMamba agreement was 78.4%, manual versus SegmentAnyTree 65.8%, manual versus ForestMamba+BluePoint 72.1% and ForestMamba versus ForestMamba+BluePoint 77.5%. At Mathislewald, manual versus ForestMamba and manual versus SegmentAnyTree were each 81.3%, manual versus ForestMamba+BluePoint was 65.6% and ForestMamba versus ForestMamba+BluePoint was 78.1%.

Table 11: Stem-level DetailView species-label agreement (%) between instance tracks.

Pair	Ettenheim ($N = 111$)	Mathislewald ($N = 32$)
Manual vs SegmentAnyTree	65.8	81.3
Manual vs ForestMamba	78.4	81.3
Manual vs ForestMamba+BluePoint	72.1	65.6
ForestMamba vs ForestMamba+BluePoint	77.5	78.1

Row-normalized confusion matrices for all four tracks on each site are shown in Figures A.13 and A.14. Relative to the manual panels, automatic tracks redistribute residual mass among open-set and neighbouring taxa while preserving the same true-class supports. Per-class F1 for inventory-supported taxa under each track is summarised in Figure 24. At Ettenheim, Douglas fir F1 remained comparatively high across tracks, whereas silver fir and European beech F1 declined under ForestMamba and ForestMamba+BluePoint relative to the manual track. At Mathislewald, Norway spruce F1 stayed high under SegmentAnyTree and declined under ForestMamba+BluePoint. European beech and silver fir F1 varied with instance source on the smaller supports.

**Figure 24:** DetailView per-class F1 by instance source for inventory-supported taxa at Ettenheim and Mathislewald.

6 Discussion

6.1 Overall performance

Overall performance covers plot-level instance recovery, detection versus delineation diagnostics, and DetailView accuracy on inventory-matched stems under manual instances. Structural associations and segmentation-to-species propagation are treated in the following subsections.

6.1.1 Reference extent and inventory matching

The Ettenheim reference comprises four completed 25 m subplots (approximately 50 m × 50 m) with 128 manual ground-truth trees on 26 997 057 evaluation points (22 714 670 tree points). Per-subplot counts of 25, 41, 38 and 24 indicate uneven stocking, with tiles 002 and 003 carrying the

largest tree loads. Mathislewald contributes one annotated subplot (approximately $25\text{ m} \times 25\text{ m}$) with 189 ground-truth trees on 24 260 295 evaluation points (14 605 364 tree points), counting each retained manual CloudCompare tree layer separately. Tree count per unit area and tree-point density are both higher than on any Ettenheim tile, consistent with a more crowded proximal scene.

These N_{GT} values define the denominators for recall and related detection metrics and are distinct from the inventory-matched stems used for species classification (Figure 9). Exclusive assignment matched 111 of 115 inventory stems at Ettenheim (4 unmatched) and 32 of 34 at Mathislewald (2 unmatched), while 17 and 157 manual layers remained unmatched. At Ettenheim the census is nearly complete: almost every inventoried stem maps to a unique manual layer, with only a small residual for review. At Mathislewald almost every inventory stem is matched, but the large unmatched-layer residual (157 of 189) reflects the many manual trees without an exclusive inventory partner (small stems, stubs and other non-inventory residuals) relative to a short inventory of 34 stems rather than 157 omitted field trees. Inventory matching therefore selects the taxonomic evaluation set. It does not redefine the geometric reference.

Qualitative instance colourings place ForestMamba and SegmentAnyTree on the same point support as the manual reference, so metric differences reflect label assignment rather than changes in geometry. At Mathislewald that shared support makes the detection shortfall visible as many manually coloured crowns without a corresponding automatic instance. Breast-height inventory overlays show markers both coincident with and offset from breast-height layer centroids. Residual offsets motivate the visual co-registration step and contribute to assignment uncertainty noted under Limitations.

6.1.2 Plot-level performance

At $\tau = 0.5$, ForestMamba achieved high recall at Ettenheim (91.4%) but only 26.5% recall at Mathislewald against the full 189-tree reference, still exceeding SegmentAnyTree (75.8%; 10.6%). Coverage ranked identically. Precision diverged by site: ForestMamba over-predicted at Ettenheim ($N_{\text{pred}} = 205$ versus 128 ground-truth trees; precision 57.1%) and kept moderate precision at Mathislewald (61.7%) while missing most reference trees (139 false negatives among 81 predictions). SegmentAnyTree retained moderate precision at Ettenheim (55.7%) and performed poorly at Mathislewald (23.3%), with 66 false positives and 169 false negatives among 86 predictions. Plot-level F1 favoured ForestMamba on both sites under this reference (70.3% versus 64.2% at Ettenheim; 37.0% versus 14.5% at Mathislewald). The between-model gap remains clear in the denser spruce subplot, but absolute Mathislewald scores are low for both models once every split manual tree layer is scored as its own ground-truth object.

ForestMamba therefore operates as a high-recall detector at Ettenheim whose Mathislewald recall collapses when many small or separately delineated reference trees must each be recovered as their own object, whereas SegmentAnyTree is competitive at Ettenheim but highly sensitive to local packing and product alignment at Mathislewald. Final SegmentAnyTree labels used absolute-coordinate partner products rather than earlier height-normalized transfers. The Mathislewald

result accordingly reflects both model behaviour and product preparation under the shared evaluation protocol.

6.1.3 Ettenheim subplot heterogeneity

ForestMamba F1 ranged from 72.7% (tile 001) to 63.0% (tile 004), with coverage remaining high (80.3-94.3%). SegmentAnyTree F1 ranged from 65.4% (tile 002) to 46.5% (tile 003), where coverage fell to 52.9% and 15 trees were missed. ForestMamba false negatives remained low across tiles (1-5). Error centroids are distributed through subplot interiors rather than confined to edges. Plot-level aggregates therefore mask tile-level failure modes that would matter under operational acceptance thresholds.

6.1.4 IoU threshold dependence

Across $\tau \in \{0.3, \dots, 0.7\}$, ForestMamba F1 at Ettenheim declined only from 70.9% to 66.1%, indicating that accepted matches largely exceed the default gate. SegmentAnyTree declined from 68.9% to 50.3%, indicating a larger share of marginal overlaps. At Mathislewald, ForestMamba declined from 46.7% to 23.7% while remaining ahead of SegmentAnyTree, which declined from 26.2% to 7.3%. Per-tree maximum-IoU distributions agree: ForestMamba still concentrates more mass at higher best IoU than SegmentAnyTree at Mathislewald, but both leave a large share of reference trees with low or zero overlap, so increasing τ removes many of the remaining true positives.

6.1.5 Error taxonomy and spatial structure

Over- and under-segmentation counts at $\tau = 0.5$ are small relative to misses and leakage-flagged true positives. At Ettenheim, ForestMamba recorded 11 misses and 10 leakage-flagged matches. SegmentAnyTree recorded 31 and 42. At Mathislewald, ForestMamba recorded 139 misses and 18 leakage-flagged matches. SegmentAnyTree recorded 169 misses and 12 leakage-flagged matches. Leakage-flagged matches pass the IoU gate while sharing substantial point mass with neighbouring trees or background and therefore indicate delineation defects among detections. SegmentAnyTree’s elevated leakage-flag rate at Ettenheim, despite usable F1, shows that many accepted objects remain geometrically impure. At Mathislewald the dominant failure for both models is simply missing most reference trees rather than impure matches alone: the corresponding appendix plan views (Figure A.9) show false-negative centroids throughout the subplot, not a narrow edge band of omissions.

6.1.6 Plot F1 versus macro hard F1

Macro hard F1 weights each ground-truth tree equally. Plot F1 is dominated by the global TP/FP/FN pool. At Ettenheim, ForestMamba macro hard F1 (88.3%; 95% CI 83.2-92.9) exceeds plot F1 (70.3%), consistent with strong delineation among detected trees and a large false-positive penalty at plot scale. SegmentAnyTree values are closer (67.5% versus 64.2%). At Mathislewald, ForestMamba macro hard F1 (22.8%; 95% CI 17.3-28.4) sits below plot F1 (37.0%), consistent with many zero-scoring missed trees pulling the per-tree mean down while the fewer matched

trees still contribute to the pooled TP/FP/FN score. SegmentAnyTree remains low on both (8.7% versus 14.5%). The Ettenheim ForestMamba contrast shows that a detection-heavy error mode can coexist with high per-tree overlap once a match exists. The Mathislewald contrast shows the opposite imbalance when misses dominate.

Against that backdrop, under the shared protocol ForestMamba scores higher on both sites on this reference. Because editing began from preliminary ForestMamba instances, that ranking is conditional on the reference construction and should not be read as an unconditional claim of superiority over SegmentAnyTree on independently delineated ground truth. The models differ in detection versus delineation behaviour: ForestMamba is high-recall and precision-limited at Ettenheim and miss-dominated at Mathislewald. SegmentAnyTree is more threshold-sensitive and fails under adverse packing or small-object occupancy. Those differences affect any inventory that assumes one predicted object per physical tree and are consistent with the method- and site-dependent spreads reported in recent TLS and multi-platform benchmarks (Cherlet et al., 2026; Henrich et al., 2024; Wang et al., 2016; Wielgosz et al., 2024; Xiang et al., 2024).

6.1.7 DetailView on manual instances

DetailView was evaluated on inventory-matched stems only ($N = 111$ Ettenheim; $N = 32$ Mathislewald). Class supports are imbalanced: Ettenheim comprises 79 European beech, 17 silver fir, 13 Douglas fir, 1 European larch and 1 Norway spruce; Mathislewald comprises 21 Norway spruce, 7 European beech and 4 silver fir. Bootstrap intervals for overall accuracy are wide, particularly at Mathislewald (46.8-78.1% on the manual track). Species metrics therefore apply to an inventory-filtered subset of the geometric reference.

On manual instances, overall accuracy was 46.8% at Ettenheim and 62.5% at Mathislewald (macro F1 36.1% and 68.7%; pooled OA 50.3%). These values remain well below published FOR-species20K test performance despite use of manually refined instances (Puliti et al., 2025). Taxonomic transfer, class imbalance and open-set predictions therefore dominate the error budget prior to any change of instance source, in line with long-standing challenges for operational tree-species mapping from remote sensing (Fassnacht et al., 2016; Mustafa and Habeeb, 2014; Pu, 2021).

Confusion matrices locate the errors. At Ettenheim, Douglas fir reached about 92% on the diagonal, whereas European beech reached only about 42% despite majority support. Silver fir was split mainly between correct silver fir (about 41%) and Douglas fir predictions (about 35%). The single European larch stem and the single Norway spruce stem were each predicted as Douglas fir. Beech misclassification is the principal local failure, consistent with application of a 33-class European head without local recalibration, including open-set taxa absent from the plot label set and conifer confusion among related softwoods. At Mathislewald, diagonals were higher for Norway spruce (about 67%) and moderate for beech and silver fir (about 57% and 50%), with residual mass on neighbouring labels. With $N = 32$, off-diagonal cells represent few stems and should be treated as directional rather than precise rates. Higher overall accuracy in the spruce-dominated subplot is consistent with stronger recovery of the local majority conifer

and a smaller broadleaf share.

Under manual instances, DetailView recovers inventory-compatible labels only moderately well. Accuracy is site- and class-dependent and is limited by open-set transfer into a small local taxonomy. Geometric instance quality is not the primary constraint on this baseline.

6.2 Structural relationships

Structural covariates separate site-specific failure regimes for segmentation and organise DetailView correctness differently from geometric hard F1.

6.2.1 Structural failure regimes

Structural covariates separate two failure regimes under exploratory two-sided Spearman tests (effect sizes emphasised; uncorrected p -values and Benjamini–Hochberg q within site reported for transparency). Associations with no clear evidence under $p < 0.05$ are retained below with their ρ and p . At Ettenheim, local packing is the strongest continuous associate of per-tree hard F1: Spearman ρ with nearest-neighbour distance between whole-tree centroids is 0.28 ($p = 0.0016$, $q = 0.0047$) for ForestMamba and 0.41 ($p < 0.001$, $q < 0.001$) for SegmentAnyTree, and both remain positive after residualising on point count. Crowded trees (low NN) lose roughly 22 percentage points of ForestMamba hard F1 and 45 points of SegmentAnyTree hard F1 relative to the most open tercile. That pattern matches the expectation that interlocking crowns and short gaps between neighbouring whole-tree centroids increase boundary leakage and missed detections. Height and whole-tree convex-hull volume move in the same direction as size, but ForestMamba height ($\rho = 0.04$, $p = 0.68$) and both models’ hull-volume correlations (ForestMamba $\rho = 0.08$, $p = 0.36$; SegmentAnyTree $\rho = 0.17$, $p = 0.052$) show no clear evidence under $p < 0.05$. ForestMamba in particular already scores highly across much of the height range, so a height cutoff alone would discard little of its residual error. At Mathislewald the ranking partly reverses toward size. Height and whole-tree convex-hull volume dominate (ForestMamba $\rho = 0.57$, $p < 0.001$ and $\rho = 0.64$, $p < 0.001$; SegmentAnyTree $\rho = 0.42$, $p < 0.001$ and $\rho = 0.44$, $p < 0.001$), while nearest-neighbour distance also meets $p < 0.05$ for ForestMamba ($\rho = 0.31$, $p < 0.001$) but shows no clear evidence for SegmentAnyTree ($\rho = 0.07$, $p = 0.35$). Absolute scores remain low: small, low-volume objects are almost never recovered (lowest height and hull terciles near 0–4% hard F1) while larger trees partially recover (upper height terciles 53.1% ForestMamba and 21.2% SegmentAnyTree). Failures are therefore not confined to a single structural niche. Ettenheim errors concentrate where neighbours are close, whereas Mathislewald errors concentrate on small occupancy across both models, with ForestMamba still ahead of SegmentAnyTree on the larger objects.

6.2.2 DetailView structural gradients

Height stratification produced little change in species OA: stems taller than 5 m yielded 46.3% OA at Ettenheim ($N = 108$) and 68.2% at Mathislewald ($N = 22$), and Spearman links between height and correctness showed no clear evidence of association under $p < 0.05$ (Ettenheim $\rho = -0.04$, $p = 0.66$; Mathislewald $\rho = 0.21$, $p = 0.25$). Lifeform collapse at Ettenheim showed

higher conifer than broadleaf OA (59.4%, $N = 32$ versus 41.8%, $N = 79$). The Mathislewald broadleaf subset ($N = 7$) is too small for a stable lifeform contrast. Majority broadleaf stems at Ettenheim remain the hardest taxonomic target. Height filtering alone does not restore species performance on the matched set.

Nearest-neighbour packing among whole-tree centroids and whole-tree convex-hull volume tell a different story from the segmentation analysis. At Ettenheim, open neighbourhoods only weakly improve species OA (42.1% crowded versus 56.8% open on the manual track; $\rho = 0.12$, $p = 0.21$), so there is no clear evidence that free space acts as a taxonomic quality filter under $p < 0.05$ even though it improves geometric F1. Whole-tree convex-hull volume instead shows a clear inverse association under $p < 0.05$ (manual OA 64.9% / 45.9% / 29.7% from low to high volume; $\rho = -0.28$, $p = 0.003$). Large-occupancy matched stems at Ettenheim are dominated by European beech, the class with the weakest diagonal in the confusion matrix. The hull-volume gradient therefore largely rediscovers the beech failure rather than a pure geometric effect of whole-tree bulk. ForestMamba and ForestMamba+BluePoint reproduce the same decline, which indicates that the pattern is taxonomic rather than an artefact of one delineation source.

At Mathislewald, open neighbourhoods show higher manual OA than crowded ones in this recount (36.4% crowded versus 70.0% open; $\rho = 0.24$, $p = 0.18$), and whole-tree hull volume remains without clear rank association (manual $\rho = 0.17$, $p = 0.36$). With $N = 32$ and roughly ten stems per bin, that pattern should not be over-interpreted, but it warns against treating Mathislewald species strata as interchangeable with Ettenheim packing rules.

Overall, Ettenheim residual geometric error concentrates where whole-tree centroids are tightly packed, whereas Mathislewald residual geometric error concentrates on small occupancy. Structural covariates useful for segmentation (packing at Ettenheim; size at Mathislewald) are not interchangeable predictors of DetailView correctness.

6.3 Segmentation-to-species effects

The same inventory-matched stems were classified under manual, SegmentAnyTree, ForestMamba and ForestMamba+BluePoint instances, isolating instance source as the experimental factor. Overall accuracy therefore conditions on that matched-stem set: trees missed by automatic segmentation outside the set, and unmatched manual layers, do not enter the score. The ablation isolates label propagation given instances rather than end-to-end species-map performance.

At Ettenheim, overall accuracy declined by 0.9 percentage points under SegmentAnyTree, by 1.8 under ForestMamba and by 3.6 under ForestMamba+BluePoint relative to the manual track. Paired bootstrap 95% intervals for those Δ OA values all include zero, so the Ettenheim gaps are compatible with sampling noise on the matched stems. At Mathislewald, SegmentAnyTree exceeded the manual track by 6.3 percentage points, while ForestMamba and ForestMamba+BluePoint fell short by 6.3 and 12.5 points. Paired intervals are wide ($N = 32$) and do not support a strong causal ranking of instance tracks. BluePoint did not improve species OA relative to ForestMamba (-1.8 points at Ettenheim; -6.3 at Mathislewald) and reduced macro F1 on both sites.

Stem-level label agreement between tracks was high for manual versus ForestMamba (78.4% Ettenheim; 81.3% Mathislewald), lower for manual versus SegmentAnyTree at Ettenheim (65.8%; 81.3% at Mathislewald), intermediate for manual versus ForestMamba+BluePoint (72.1%; 65.6%), and high for ForestMamba versus ForestMamba+BluePoint (77.5%; 78.1%). Combined with modest OA gaps, high agreement on several pairs indicates that many stems retain the same label (correct or incorrect) across instance sources. Segmentation error reassigns a minority of decisions rather than rewriting the taxonomic map.

Multi-track confusion matrices preserve true-class supports and redistribute residual mass among open-set and neighbouring taxa. At Ettenheim, Douglas fir remains comparatively stable across tracks, while beech and silver fir absorb more automatic-track degradation. Per-class F1 declines accordingly under ForestMamba and ForestMamba+BluePoint. At Mathislewald, Norway spruce F1 remains high under SegmentAnyTree and weakens under ForestMamba+BluePoint. Beech and silver fir fluctuate on small supports. Class-level shifts can therefore occur even when plot-level OA appears nearly flat.

Segmentation error propagates into stem-level species labels but is not the dominant barrier in this study. Ettenheim manual-to-automatic OA gaps remain within a few percentage points against a manual baseline near 47%. Mathislewald signed gaps are larger in magnitude but rest on a sample too small for strong causal ranking of instance tracks. Closing the taxonomic gap on these matched stems is unlikely without domain adaptation, confidence gating, restriction to local taxa or a larger matched sample (Fassnacht et al., 2016; Hamedianfar et al., 2022; Puliti et al., 2025).

6.4 Limitations

Manual delineation was performed without peer review or double annotation. Thin branches, interlocking crowns and suppressed stems remain ambiguous, and residual layer errors can persist when a reference is built by one person (Burt et al., 2019; Cherlet et al., 2026; Wilkes et al., 2017). An independent re-annotation of a shared subsample would quantify reference uncertainty. Absent that control, model scores are distances to this delineation rather than to a consensus reference.

Inventory assignment retained a visual component. Geometric rules and attribute tie-breaks handled most pairs, but overlays, a translational shift at Mathislewald and a small override list were used where markers were offset from breast-height layer centroids. Visual judgement introduces additional error where neighbouring stems have similar DBH or height (Cavalheiro et al., 2026; Kükenbrink et al., 2022; Liu et al., 2018). Exclusive one-to-one matching also approximates a stem census that inventory and TLS may define inconsistently.

Time constraints limited spatial coverage: not all clipped subplots were annotated, so the analysed footprints are smaller than the available archives and less representative of stand-level variability. The manual reference was also not constructed from an unlabeled cloud. Editing began from preliminary ForestMamba instances with subsequent splits, merges and cleaning. This accelerated reference building but leaves a residual risk that the reference inherits ForestMamba

inductive structure despite major correction. Bias of that form would tend to favour ForestMamba on ambiguous crowns and interlocking boundaries that were resolved during editing in ways consistent with the seed labels, while within-protocol comparisons and SegmentAnyTree product caveats at Mathislewald remain informative. Model ranking for overall instance performance is therefore conditional on this reference rather than an unconditional claim against independently delineated ground truth (Cherlet et al., 2026; Henrich et al., 2024).

Species evaluation used inventory-matched stems only. Unmatched manual trees do not enter DetailView scoring. Results therefore condition on co-registration and matching success and inherit inventory omission, positional noise and the local species list (Fassnacht et al., 2016; Puliti et al., 2025). They do not fold missed or extra automatic trees outside the matched set into a species-map score. DetailView was applied as an off-the-shelf European FOR-species20K model without retraining or label-set restriction, so open-set predictions are expected. BluePoint was tested only as a ForestMamba post-refinement on matched stems. Each site used a single TLS modality despite ULS and related platforms in the broader ECOSENSE and WaldAgil portfolios. Cross-platform splits were out of scope (Guimarães et al., 2020; Puliti et al., 2025; White et al., 2016; Wielgosz et al., 2024). ForestFormer3D supplied the evaluation template but was not run as a predictor (Xiang et al., 2025), and no end-to-end fine-tuning was performed. Bootstrap intervals capture sampling uncertainty on matched stems but not reference-label uncertainty or spatial autocorrelation within contiguous subplots. Per-tree Spearman associations therefore describe structure within the annotated footprints rather than independent plot replicates. A fixed-radius neighbour count with the same edge rule was examined as a denser packing descriptor but did not clearly add beyond nearest-neighbour distance once residualised on that spacing measure, and edge censoring left only a minority of trees in the interior set. Planimetric whole-tree hull overlap on those same interior focals was checked as a more geometric interlocking descriptor but likewise did not clearly add beyond nearest-neighbour distance after residualising on spacing, and at Ettenheim it tracked whole-tree hull volume closely enough that it was not treated as an independent packing axis.

6.5 Scope and generalizability

The study is a local benchmark on two university research forests in southwestern Germany: a larger beech-dominated mixed stand at Ettenheim and a smaller spruce-dominated subplot at Mathislewald. Extrapolation to boreal plantations, tropical forests, leaf-off deciduous scenes or airborne-only inventories requires new references. Within temperate mixed TLS settings, the results are most informative for relative model behaviour under a shared protocol, for the magnitude of segmentation-to-species propagation on inventory-matched stems, and for the cost of building usable geometric and taxonomic references. Two ecologically distinct sites strengthen external validity. Small matched N at Mathislewald, partial subplot coverage and the lack of an independent re-annotation weaken it.

6.6 Implications

The practical reading across the three question blocks is therefore structural triage for QA, not a single preferred model: overall scores situate the models, structural covariates show where they fail, and segmentation-to-species contrasts show that instance source is secondary on matched stems. Nearest-neighbour packing among whole-tree centroids is a useful Ettenheim QA cue for geometric quality. At Mathislewald, object size and whole-tree hull volume are more informative than residual centroid spacing. For detection-oriented use, ForestMamba is viable at plot scale on the Ettenheim scenes if false positives are acceptable or filterable, but Mathislewald recall against the full 189-tree reference shows that many small trees remain undetected. For delineation-sensitive products (crown competition, biomass allometry, clean stem extraction), leakage-flag and τ -sweep diagnostics are as informative as headline F1. SegmentAnyTree remains a relevant Ettenheim baseline but requires careful absolute-coordinate product preparation and is not competitive on the denser Mathislewald reference under this protocol.

Off-the-shelf DetailView is not a drop-in replacement for field taxonomy on these forests. Constrained local label sets, fine-tuning or human oversight remain necessary before automated species layers support management or ecological inference. Because instance-source effects on stem-level labels are modest relative to the manual-instance taxonomic gap, and because free-space and whole-tree hull-volume filters do not recover beech performance at Ettenheim, investment in segmentation or packing-based stem triage alone will not close that taxonomic gap on matched stems. The paired manual-plus-inventory reference and dual instance/species protocol remain a reusable template for research forests that already hold TLS and stem maps.

6.7 Future research

Priority extensions are: (i) complete annotation of remaining clipped subplots and additional stands, with an independent re-annotation control sample; (ii) enlarge the inventory-matched species set and test DetailView variants restricted or recalibrated to local taxa, with confidence gating and explicit open-set handling (Fassnacht et al., 2016; Puliti et al., 2025); (iii) repeat the instance benchmark with fully independent manual delineation on a subset and with additional or fine-tuned predictors (Henrich et al., 2024; Nguyen et al., 2026; Xiang et al., 2025); (iv) add ULS or other platform tracks under the same matched stems, following related cross-platform species and instance benchmarks (Guimarães et al., 2020; Puliti et al., 2025; Wielgosz et al., 2024); and (v) release reference clouds, match manifests and evaluation scripts under a clear license. Structural DBH/height error budgets and multi-temporal revisit are natural subsequent steps once instance and species layers are stable (Calders et al., 2015; Disney et al., 2018; Kükenbrink et al., 2022; Liu et al., 2018; Pourdelan et al., 2026; Riedel and Kändler, 2017; Tian et al., 2023).

7 Conclusions

We constructed manually refined TLS instance references for Ettenheim and Mathislewald, linked inventory species where exclusive matching succeeded, and benchmarked ForestMamba and SegmentAnyTree for detection and overlap together with DetailView for species classification

on manual and automated instances. Beyond headline scores, the strongest contribution is structural: which trees and neighbourhoods remain challenging for these models.

For overall performance (RQ1), under the shared protocol and this manually refined reference ForestMamba achieved higher plot-level scores on both sites, and SegmentAnyTree remained competitive at Ettenheim but weaker at Mathislewald in the final absolute-coordinate comparison. Those rankings are conditional on a ForestMamba-seeded reference. Detection versus delineation diagnostics (threshold sweeps, leakage flags, macro hard F1) show that headline F1 alone understates how models differ, especially when misses dominate the denser subplot. On inventory-matched stems under manual instances, off-the-shelf DetailView reached only about 47% overall accuracy at Ettenheim ($N = 111$) and about three-fifths at Mathislewald. Open-set transfer and local class imbalance dominate that gap.

For structural relationships (RQ2), exploratory associations separate two site-specific failure regimes. At Ettenheim, nearest-neighbour packing among whole-tree centroids dominates per-tree hard F1, with crowded trees losing roughly 22 percentage points of ForestMamba hard F1 and 45 points of SegmentAnyTree hard F1 relative to the most open tercile. At Mathislewald, height and whole-tree convex-hull volume dominate: both models miss most small, low-volume reference trees, while larger objects only partially recover. Structural covariates useful for segmentation (packing at Ettenheim; size at Mathislewald) are not interchangeable predictors of DetailView correctness: free space showed no clear species benefit at Ettenheim, and large whole-tree hull volume there tracked poorer labels mainly through majority beech failure.

For segmentation-to-species effects (RQ3), changing the instance track on the same matched stems moved overall accuracy by only a few points at Ettenheim and by noisier but still secondary amounts at the smaller Mathislewald matched set. That ablation does not score end-to-end species maps (missed or extra trees outside the matched set are not in the OA). Better segmentation helps only modestly on these stems. It does not by itself close the taxonomic gap.

The paired geometric and inventory reference, and the dual instance/species evaluation chain, provide a reusable local benchmark template that jointly links tree size and neighbourhood structure to segmentation and species outcomes on the same stems. Priority next steps are independent re-annotation controls, larger matched samples, local species-head adaptation and cross-platform extensions using the broader ECOSENSE and WaldAgil sensing archives.

* * *

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A Supplementary diagnostic figures

This appendix collects repetitive instance-segmentation diagnostics and species panels that duplicate main-text summaries. Plot-level scores, structural hard-F1 associations (height, nearest-neighbour packing, whole-tree hull volume), and the primary DetailView species figures remain in Section 5.

A.1 Instance-segmentation diagnostics

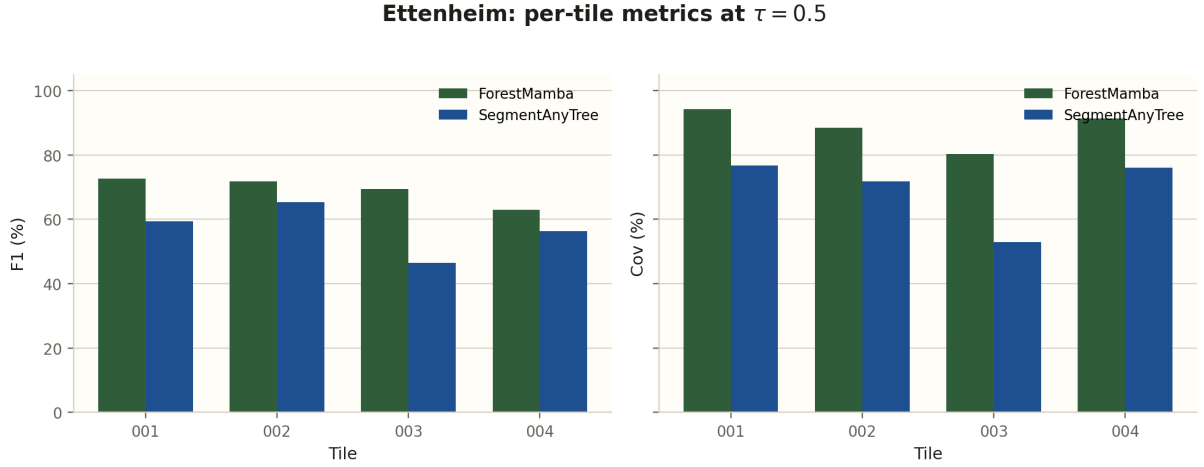


Figure A.1: Ettenheim per-subplot F1-score and coverage at $\tau = 0.5$ for ForestMamba and SegmentAnyTree.

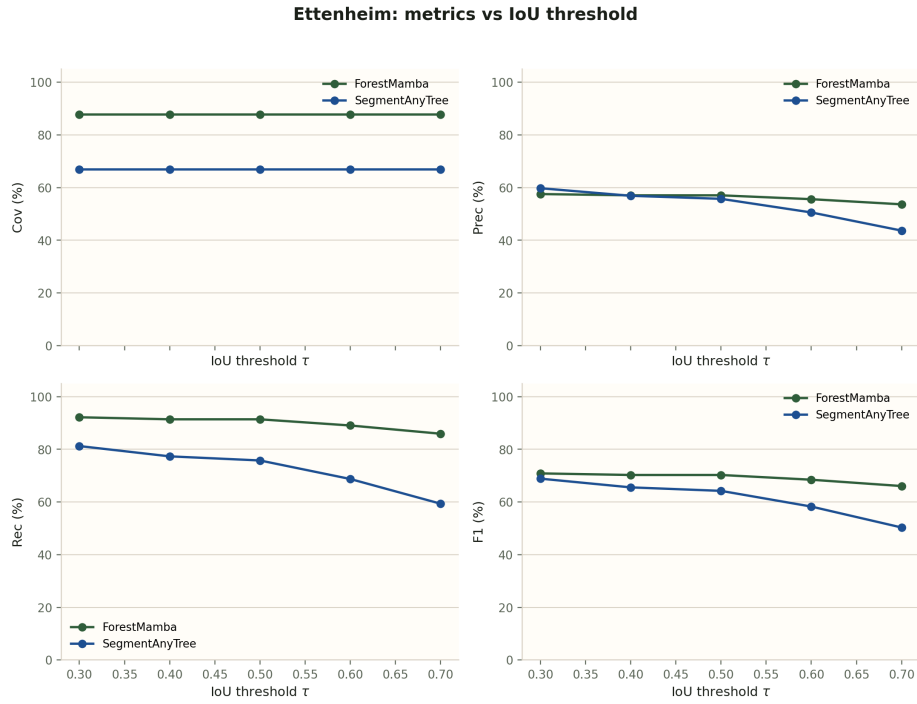


Figure A.2: Ettenheim instance metrics as a function of the IoU matching threshold τ .

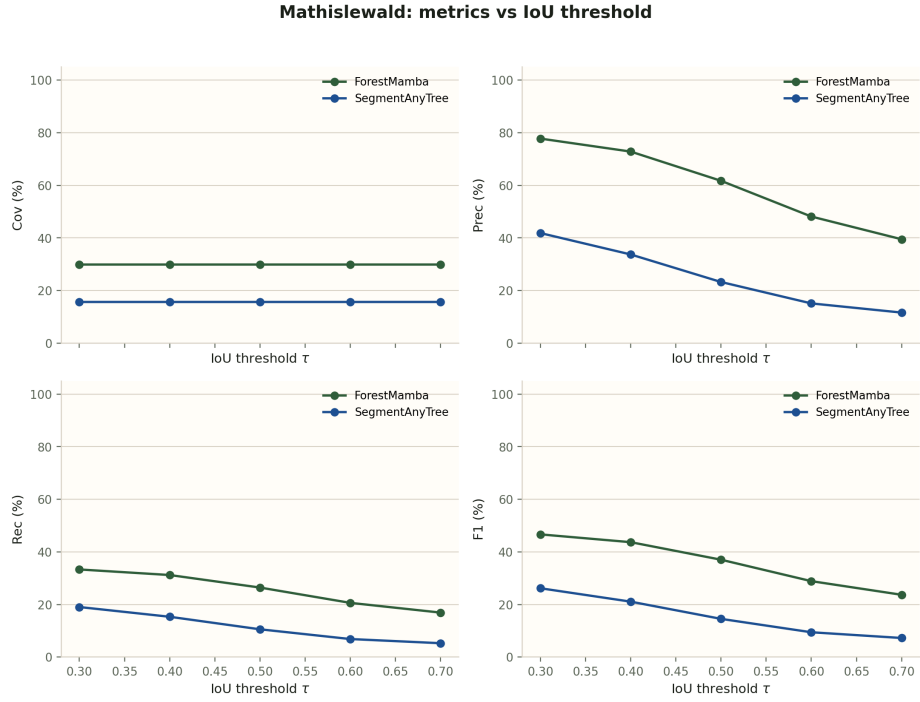


Figure A.3: MathisLewald instance metrics as a function of the IoU matching threshold τ .

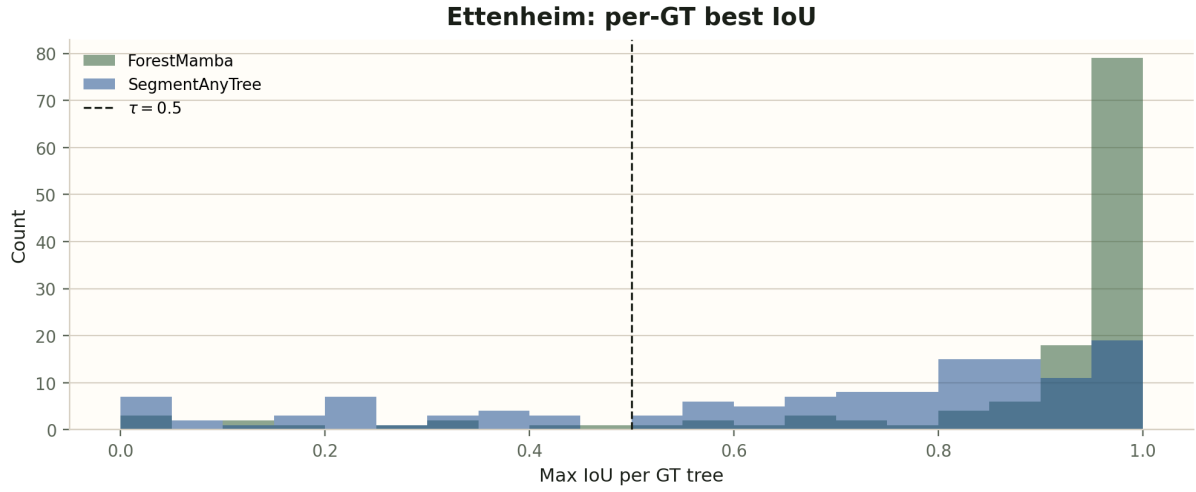


Figure A.4: Ettenheim distribution of per-ground-truth maximum IoU against ForestMamba and SegmentAnyTree.

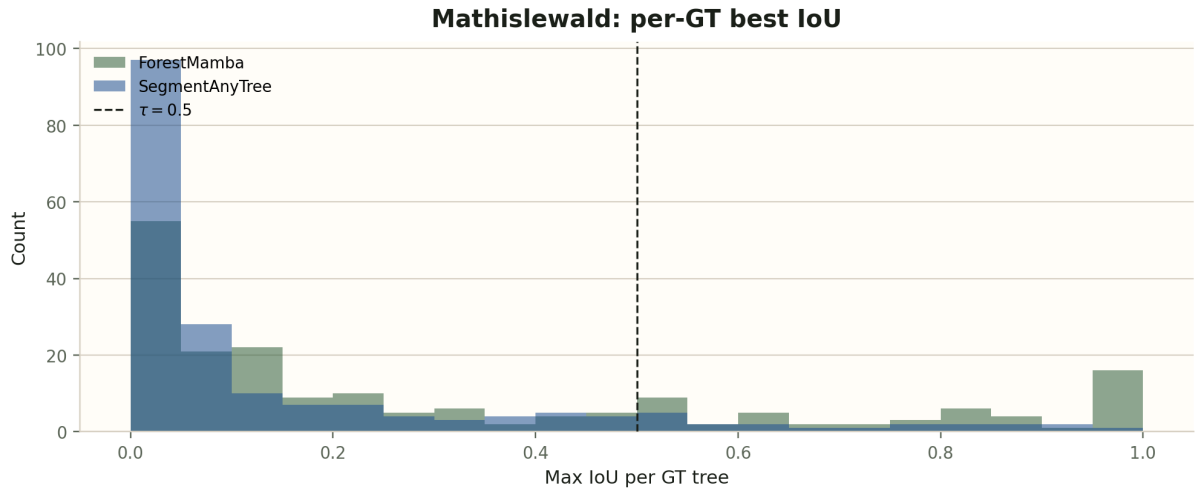


Figure A.5: Mathislewald distribution of per-ground-truth maximum IoU against ForestMamba and SegmentAnyTree.

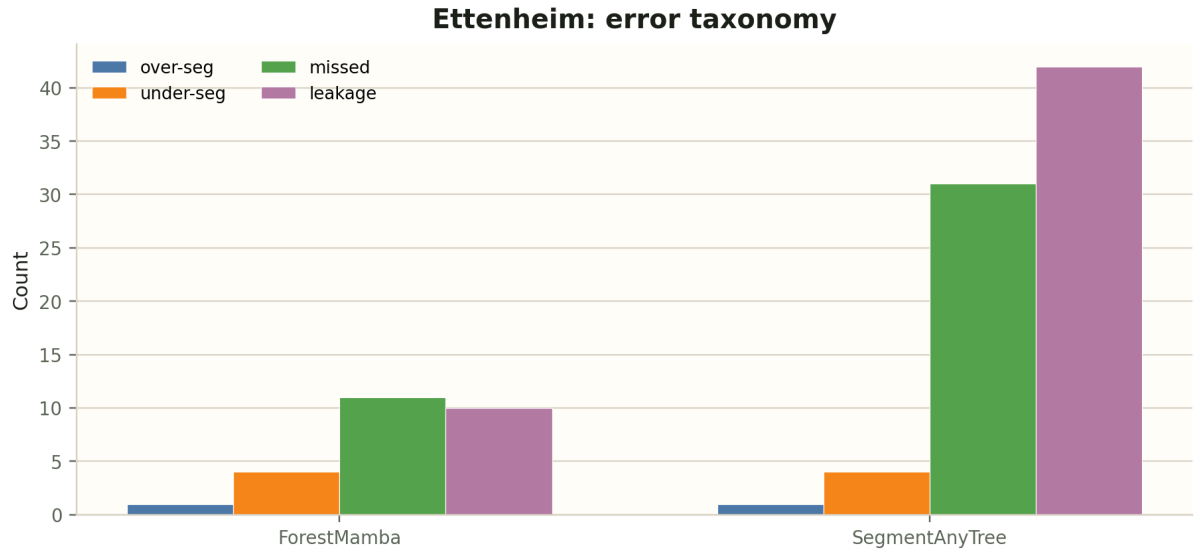


Figure A.6: Ettenheim over-segmentation, under-segmentation, missed-tree and leakage-flag counts at $\tau = 0.5$.

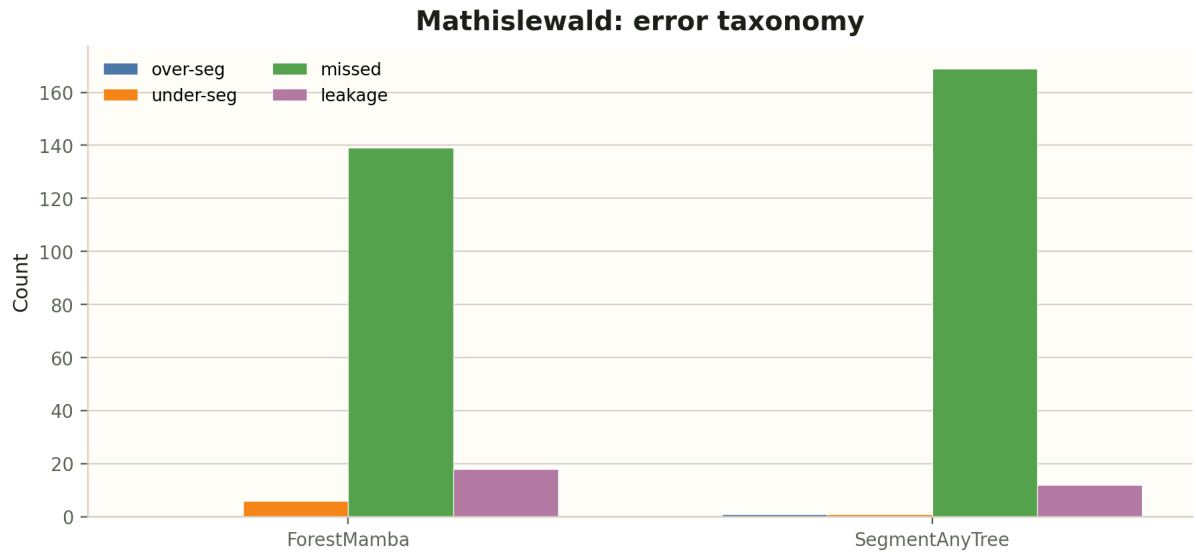


Figure A.7: Mathislewald over-segmentation, under-segmentation, missed-tree and leakage-flag counts at $\tau = 0.5$.

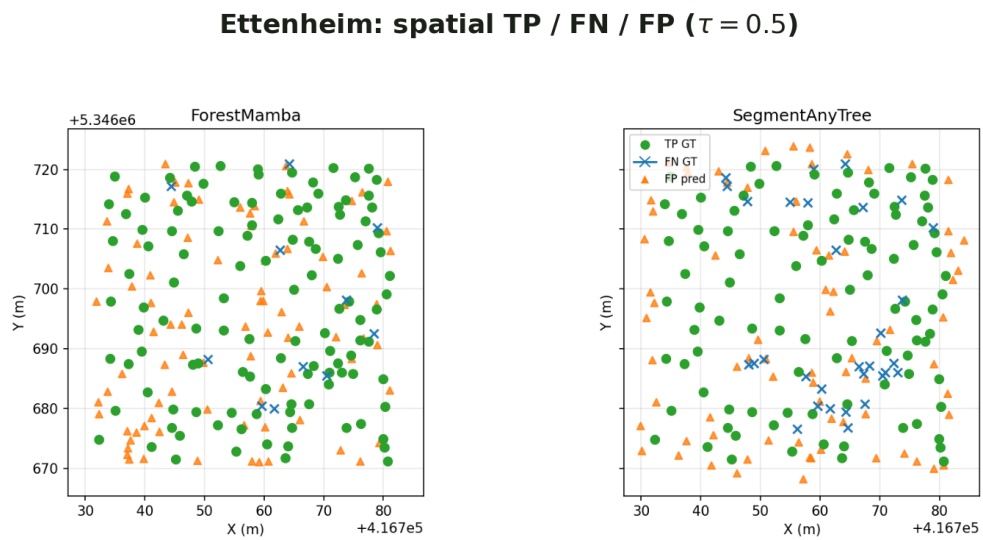


Figure A.8: Ettenheim plan view of true-positive, false-negative and false-positive instance centroids at $\tau = 0.5$.

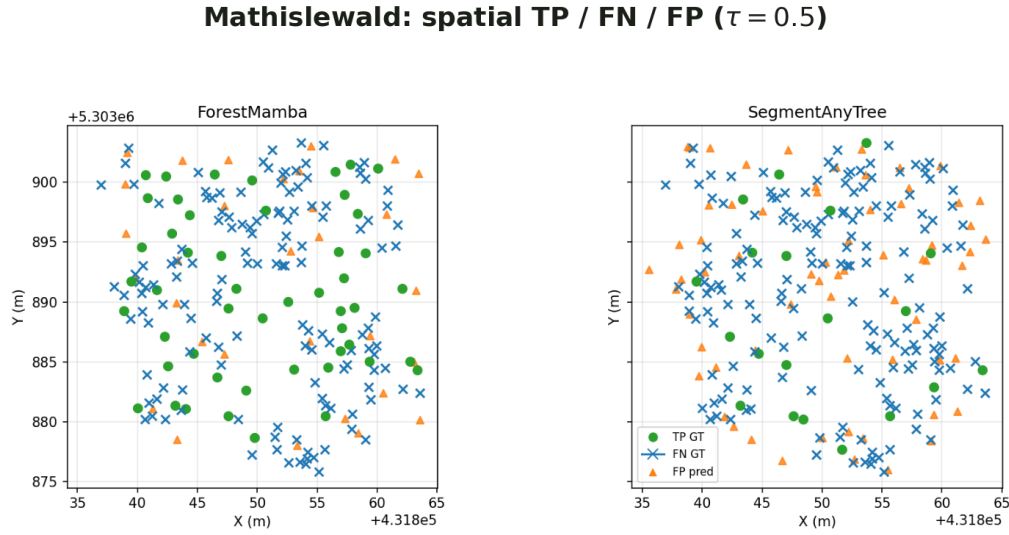


Figure A.9: Mathislewald plan view of true-positive, false-negative and false-positive instance centroids at $\tau = 0.5$. Missed ground-truth trees dominate both panels.

A.2 Species supplementary panels

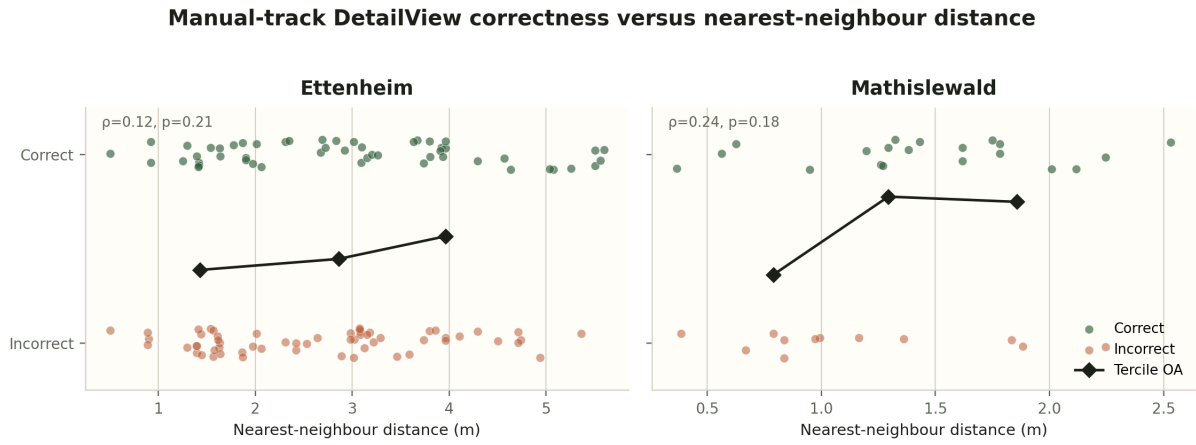


Figure A.10: Manual-track DetailView correctness versus nearest-neighbour distance between whole-tree centroids. Diamonds trace tercile overall accuracy.

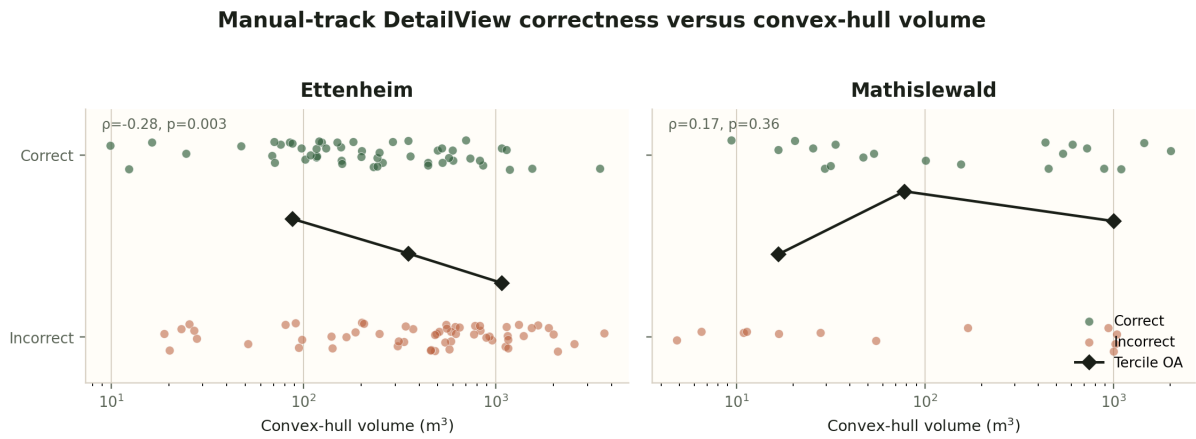


Figure A.11: Manual-track DetailView correctness versus whole-tree convex-hull volume ($\log x$). Diamonds trace tercile overall accuracy.

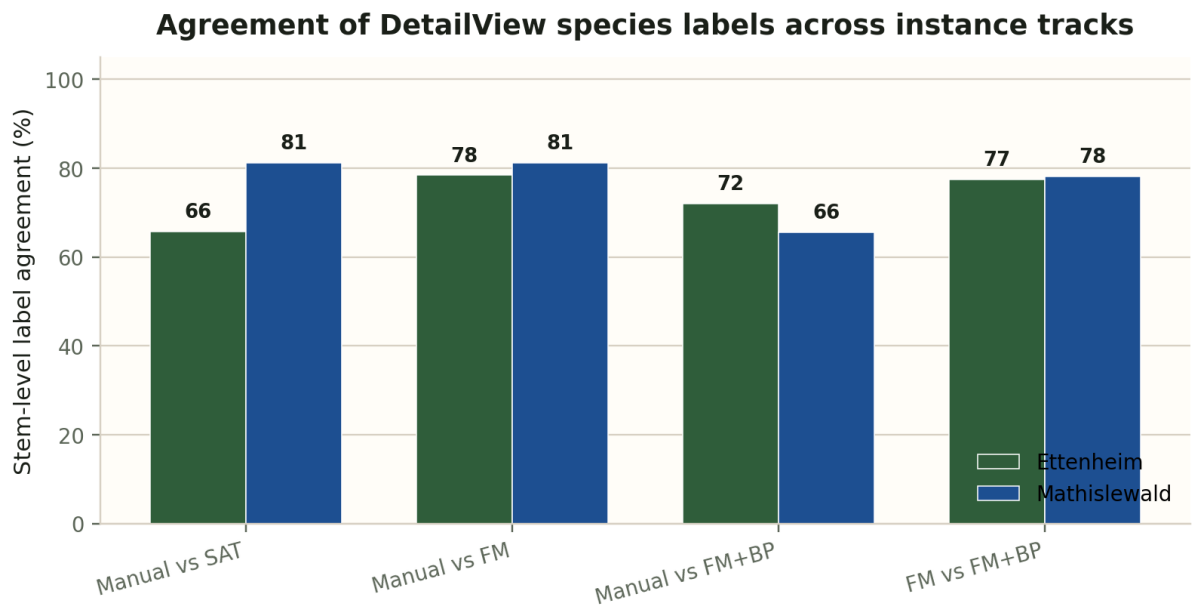


Figure A.12: Stem-level DetailView species-label agreement between instance tracks at Ettenheim and Mathislewald.

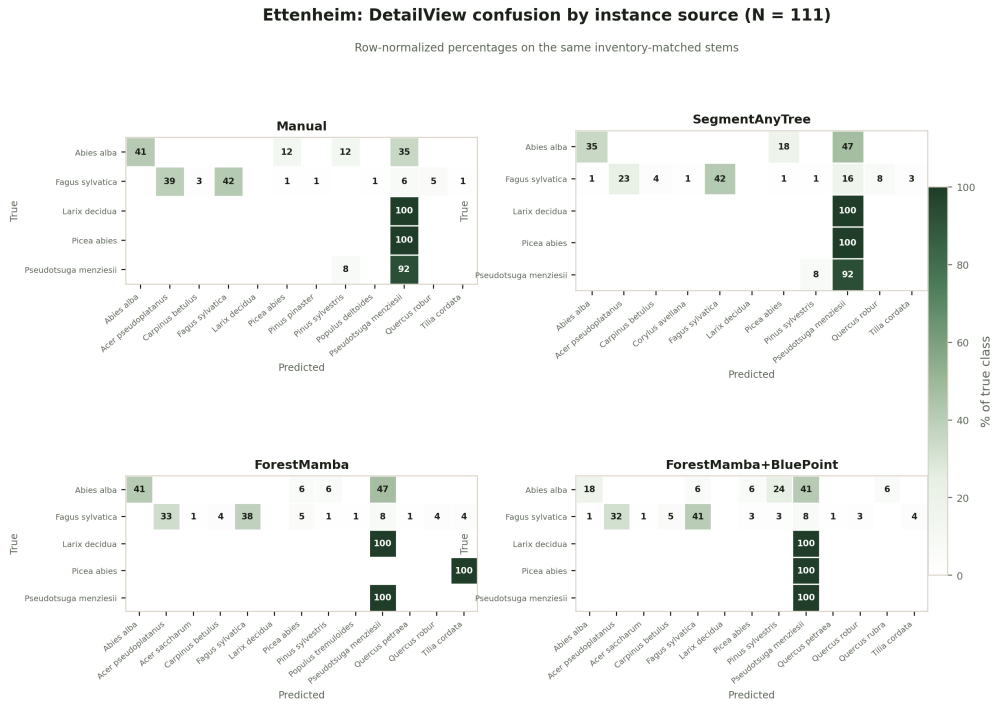


Figure A.13: Ettenheim row-normalized DetailView confusion matrices on the same inventory-matched stems under manual, SegmentAnyTree, ForestMamba and ForestMamba+BluePoint instances ($N = 111$).

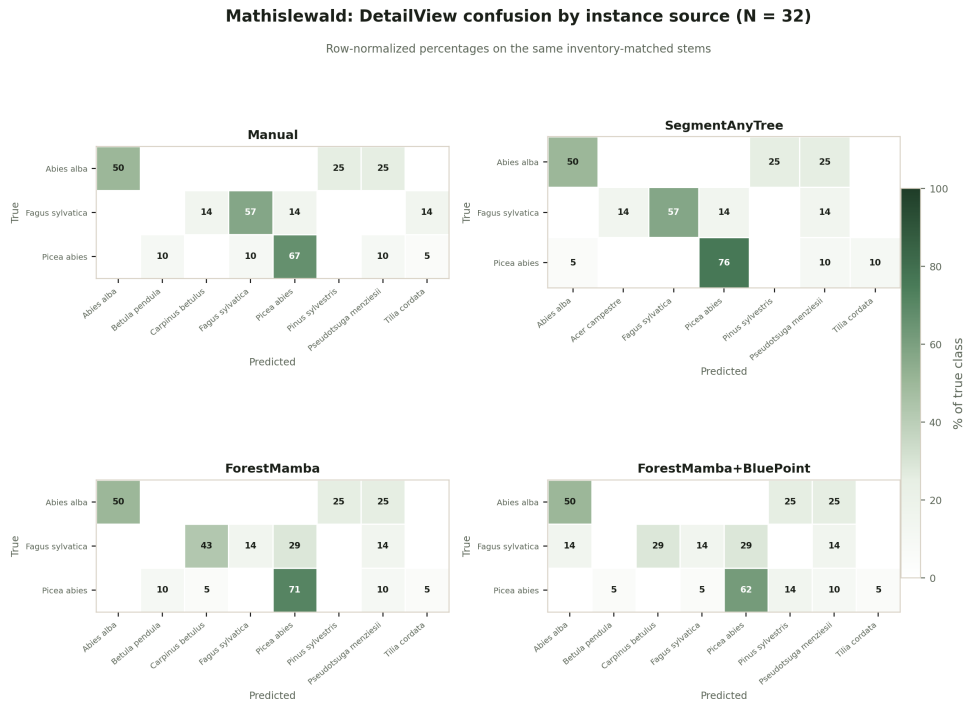


Figure A.14: Mathislewald row-normalized DetailView confusion matrices on the same inventory-matched stems under manual, SegmentAnyTree, ForestMamba and ForestMamba+BluePoint instances ($N = 32$).